



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523: DATABASE

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DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN FOR <KADA-eServe>

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NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1. Introduction

1.1 Overview about the company

The Lembaga Kemajuan Pertanian Kemubu(KADA) is a Malaysian government agency under the Ministry of Agriculture and Food Security. KADA's primary objective is to spearhead the development of the agriculture sector in the Kemubu area, a key agricultural region in Kelantan. KADA is dedicated to improving the socioeconomic status of farmers through modern farming techniques, infrastructure development, and effective management of resources.

1.2 The existing system

KADA currently relies heavily on manual, paper-based processes for member registration, loan applications, and data management. This traditional approach has led to several challenges that hinder operational efficiency and the organization's ability to adapt to modern demands.

1. Inefficiency in Member Registration and Data Entry

The manual registration process requires users to fill out paper forms, which requiring staff must then process and input into the system manually. This process is time-consuming and prone to errors, such as incorrect data input or duplication of records. These inefficiencies result in inconsistent data integrity and delays in updating the database, which affects overall system responsiveness and member onboarding timelines.

2. Limited Data Storage Capacity

KADA's current digital storage capacity for member registration is limited to on a few hundred members, posing a significant obstacle to the organization's expansion. The inability to handle a larger volume of data restricts KADA's ability to recruit and manage more members. This limitation also affects the organization's capacity to maintain comprehensive records, generate detailed reports, and conduct effective data analysis, all of which are vital for strategic planning and operational efficiency.

3. Security

The reliance on manual data entry and paper forms in KADA's current system presents significant security risks. Without proper encryption and access controls, the database is exposed to vulnerabilities, and manual entry increases the errors such as incorrect or incomplete records. These gaps compromise data confidentiality and operational efficiency.

4. Lack of Resources to Support Decision Making

The current database system lacks the features necessary to support advanced data analytics and reporting. Data is fragmented and stored in an unstructured format, hindering the ability to generate insightful reports and monitor operational trends. This limitation affects KADA's capacity to leverage historical data for strategic decision-making and hampers its ability to perform predictive analysis for future planning.

1.3 The proposed system

The system will be developed and named KADA eServe, a digital solution designed to streamline the operational process of KADA while enhancing user experience. This system has to meet the organization's needs and provides efficient management of user registration, loan applications and administrative.

To address the challenges posed by KADA's reliance on manual processes and limited digital systems, we propose the implementation of a robust, scalable and secure database-driven solution. This proposal outlines how the adoption of a database system can resolve the identified inefficiencies, improve operational efficiency and support strategic growth.

For the first problem that the company faced is addressing inefficiency in member registration and data entry. The inefficiency in member registration can be resolved by implementing an online registration system integrated with a centralized database. Members can complete digital registration forms which are validated and stored directly in the database without the need for manual data entry. Validation rules will also ensure that all inputs are consistent and error-free, reducing the chances of incorrect or duplicate records. By automating the data entry process, this solution streamlines member onboarding, making member data immediately available for other system functionalities.

Second, to overcome the limitations of KADA's current storage capacity, we have implemented a scalable relational database management system (RDBMS) such as MySQL is proposed. This will enable the system to effectively handle thousands of member records and support future expansion. Storage optimisation can be done by normalising data structures which ensuring consistency and cuts down on redundancy. Additionally, a data archiving strategy will ensure that inactive or historical records are stored securely without affecting system performance.

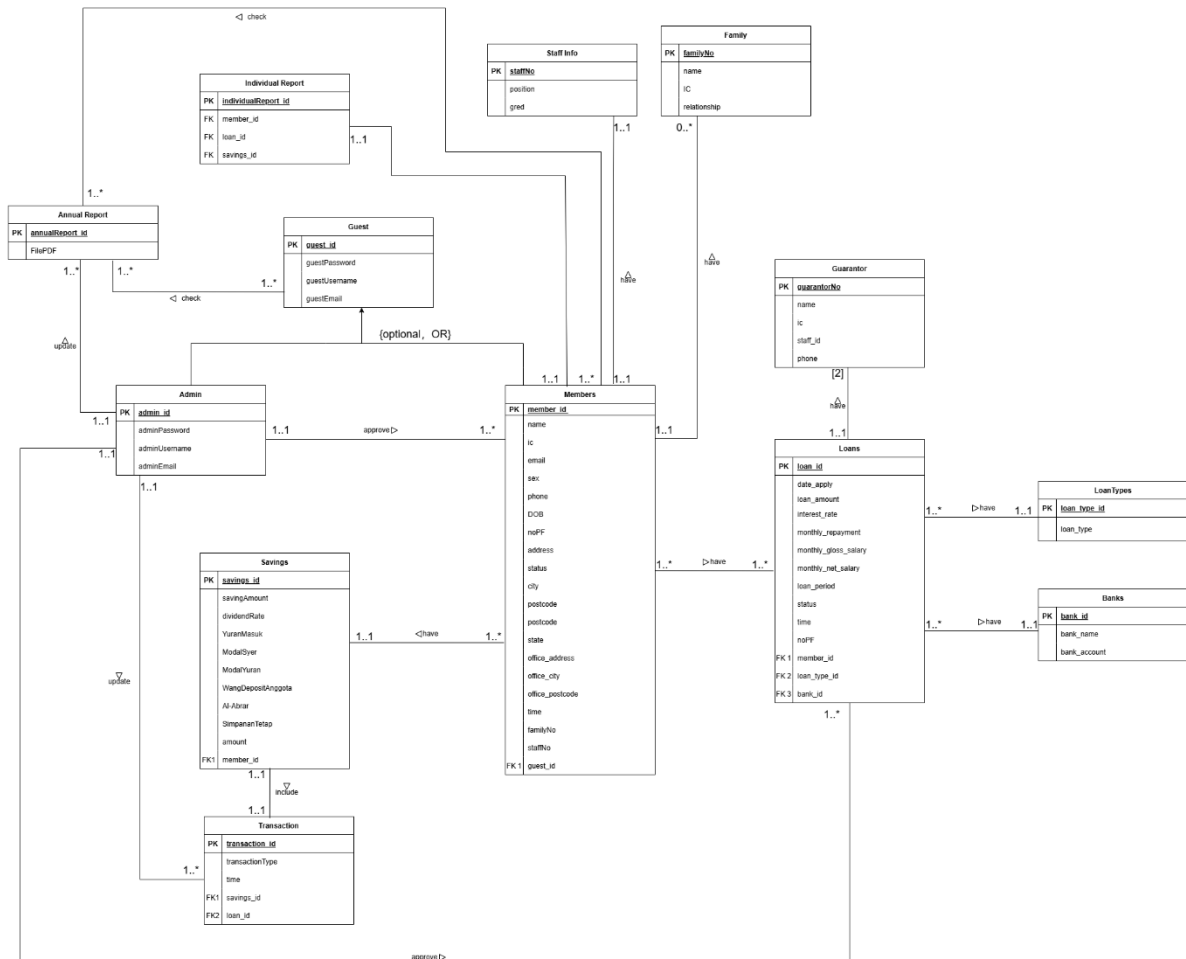
Next, the security risks associated with manual processes can be mitigated by integrating robust security mechanisms into the database system. AES and SSL/TLS encryption protocols will be used to protect sensitive data while it is being transmitted and while it is at rest. To improve confidentiality, role-based access restrictions will make sure that only individuals with permission can view or alter particular data. By taking these steps, the system's security will be greatly improved and member data will be shielded from breaches.

Lastly, a structured relational database will enable efficient querying and analysis of data. Real-time reporting will provide insights into member activities, loan statuses, and organizational performance, empowering administrators to monitor trends effectively.

In conclusion, by implementing these database solutions, KADA can transition to a modern and efficient system that overcomes existing challenges, enhances security, supports scalability and provides actionable insights for informed decision-making.

2. Global Conceptual Data Model (Global ERD)

2.1 Global Conceptual Database Design (Conceptual ERD)



2.2 Data Dictionary for global conceptual ERD

2.2.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
Guest	General term describing all users KADA-eServe	Visitor	Occurs when a user register in the system

Member	General term describing all user logged in as member using KADA-eServe	Registered staff	Occurs when a guest registration application approved by admin
Admin	General term describing admin managing KADA-eServe	Clerk, Administrator, manager	Occurs when the system built
Transaction	General term describing all transaction record key-in by admin	Deposit amount, loan repayment	Occurs when admin key-in the amount of deposit or loan repayment for each member
Savings	General term describing the savings info of each member	Yuran, Modal, Deposit, Simpanan	Occurs when a guest registration application approved by admin
Family	General term describing the family info of each member	Pewaris, Keluarga	Occurs when a guest registration application approved by admin
Staff Info	General term describing the staff position, grade info	KADA staff Info	Occurs when a guest registration application approved by admin
Guarantor	General term describing the guarantor of the member	Warrantor	Occurs when a guest registration application approved by admin
Loan	General term describing the loan info of member	Pembiayaan	Occurs when a member submits the loan application form to system
LoanTypes	General term describing the loan info of member	Loan type	Occurs when a member submits the loan application form to system

Banks	General term describing the bank account info of member	Bank account	Occurs when a member submits the loan application form to system
Individual Report	General term describing the individual report of each member saving and loan info.	Personal report	Occurs when a guest registration application approved by admin
Annual Report	General term describing the annual report of KADA	Organization report	Occurs when admin upload annual report pdf file to system

3.3.2 Identifying Relationship Types

Entity name	Multiplicity	Relationship	Entity Name	Multiplicity
Admin	1..1	update	Annual Report	1.. *
	1..1	approve	Members	1.. *
	1..1	approve	Loans	1.. *
Guest	1..1	become	Members	1..1
	1.. *	check	Annual Report	1.. *
Members	1..1	have	Savings	1..1
	1..1	have	Family	0.. *
	1..1	have	Staff Info	1..1
	1..1	apply	Loans	1.. *
	1..1	check	Individual Report	1..1
	1.. *	check	Annual Report	1.. *

Savings	1..1	include	Transaction	1.. *
Loans	1..1	have	Guarantor	[2]
	1.. *	have	LoanTypes	1..1
	1.. *	have	Banks	1..1

3.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Guest	guest_id (PK)	int	Uniquely identifies each guest	No	No	1
	guestPassword	varchar(255)	Account password for login	No	No	\$sf@142
	guestUsername	varchar(255)	Guests' full name	No	No	Ali
	guestEmail	varchar(255)	Account email for login	No	No	staff@gmail.com
	member_id (PK)	int	Uniquely identifies each member	No	No	5
	name	varchar(255)	Members' full name	No	No	Ahmad

Members	ic	int	Members' identification card number	No	No	981119-01-8732
	email	varchar(255)	Account email for receive notification	No	No	ahmad@gmail.com
	sex	varchar(255)	Members' gender	No	No	Lelaki
	phone	varchar(255)	Members' phone number	No	No	011-16596391
	DOB	date	Members' birth date	No	No	19/11/1998
	noPF	int	Members' personal profile	No	No	6
	address	text	Members' address	No	No	No 128 Taman U
	status	ENUM	Indicates the application state	No	No	Johor
	city	text	City where the member is located	No	No	Muar
	postcode	int	Postcode where the	No	No	84000

			member is located			
	state	text	State where the member is located	No	No	Kelantan
	office_address	text	Office address of member	No	No	Peti Surat 127
	office_city	text	Office city of member	No	No	Bandar Kota Bharu
	office_postcode	int	Office postcode of member	No	No	15710
	time	date	The date of submission application	No	No	05/1/2025
	familyNo	int	Family record of members	Yes	Yes	2
	staffNo	int	Staff number of member	No	No	4672
	admin_id (PK)	int	Uniquely identifies each admin	No	No	1

Admin	adminPassword	varchar(255)	Account password for admin login	No	No	\$gf@142@
	adminUsername	varchar(255)	Admins' full name	No	No	Ahmad
	adminEmail	varchar(255)	Account email for admin login	No	No	admin@gmail.com
Transaction	Transaction_id (PK)	int	Uniquely identified transaction record	No	No	6
	transactionType	ENUM	Type of transaction made	No	Yes	Loan repay, Deposit
	time	date	Date of transaction made	No	No	19/1/2025
	Savings_id (FK)	int	Indicated to the specific savings account info	No	No	7
	savings_id (PK)	int	Uniquely identified savings record	No	No	56

Savings	savingAmount	float	Amount of the member savings	No	No	3000
	dividendRate	float	Yearly dividen rate	No	No	0.03
	YuranMasuk	float	Amount of members' Yuran Masuk	No	No	2550
	ModalSyer	float	Amount of members' Modal Syer	No	No	1480
	ModalYuran	float	Amount of members' Modal Yuran	No	No	1320
	WangDepositAnggota	float	Amount of members' deposit	No	No	2550.50
	Al-Abrar	float	Amount of members' Al-Albrar	No	No	1000.50
	SimpananTetap	float	Amount of members' fixed deposit	No	No	6000.50

	amount	float	Total amount of the savings	No	No	20000
	member_id (FK1)	int	Indicated to the member who own the savings record	No	No	6
Family	FamilyNo (PK)	int	Uniquely identified family record of member	No	No	4
	name	varchar(255)	Members' family name	No	No	Ismail
	IC	varchar(255)	Members' family ic	No	No	931118-01-1236
	relationship	varchar(255)	Relationship between the family and member	No	No	Adik lelaki
Staff Info	staffNo	int	Uniquely identified each staff	No	No	9824
	position	varchar(255)	Staffs' position in KADA	No	No	perkhidmatan pelanggan

	gred	varchar(255)	Staffs' grade in KADA	No	No	M40
Guarantor	GuarantorNo (PK)	int	Uniquely identified guarantor of member	No	No	7
	name	varchar(255)	Guarantors' name	No	No	Siti
	ic	varchar(255)	Guarantors' IC	No	No	991231-01-9823
	staff_id	int	Guarantors' staff ID in KADA	No	No	7
	phone	varchar(255)	Guarantors' phone number	No	No	019-7113981
	application_id (PK)	int	Uniquely identified each application record	No	No	6
	date_apply	date	Submission date of application	No	No	20/1/2025
	loan_amount	float	Amount of loan to apply	No	No	2000.50

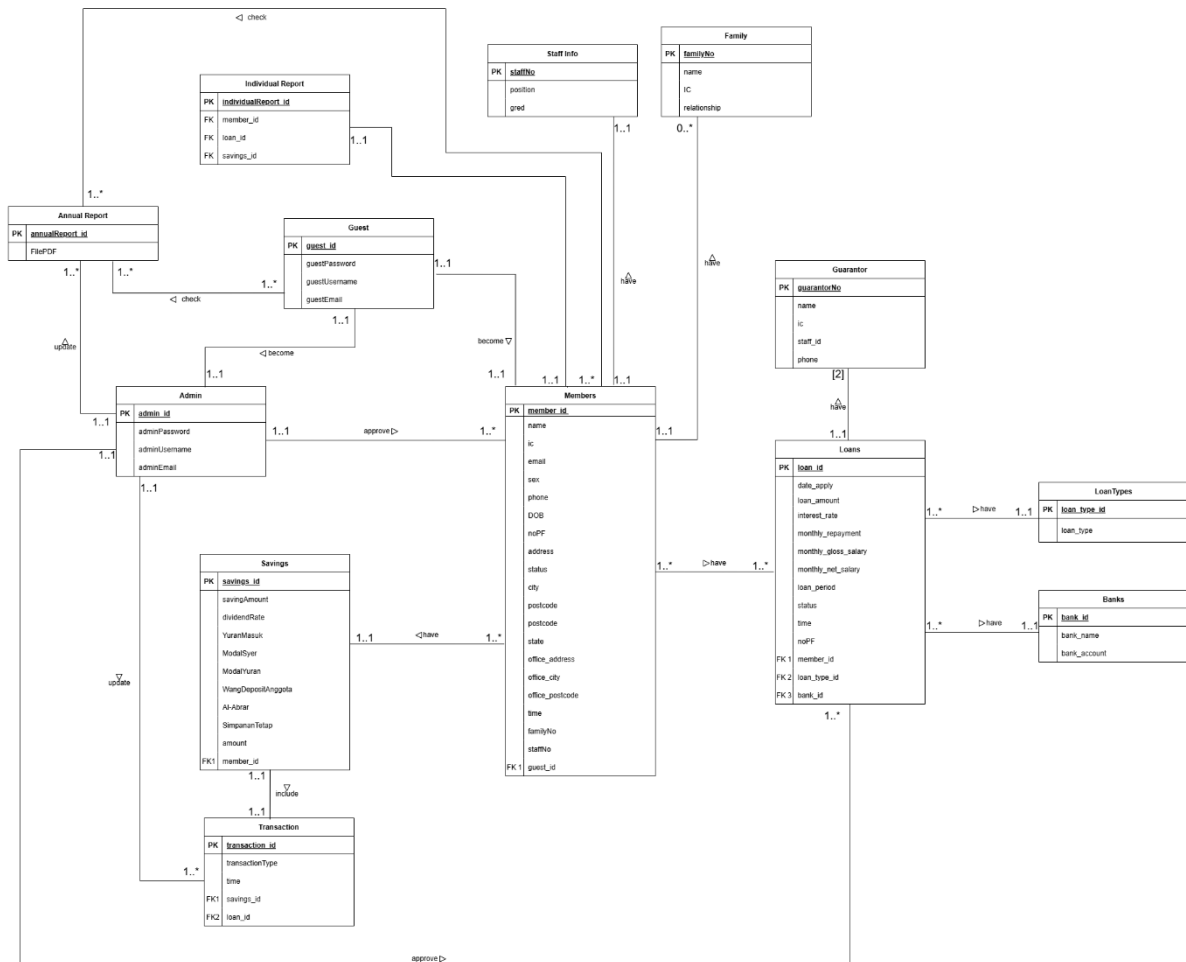
Loans	interest_rate	float	Interest rate for loan	No	No	0.05
	monthly_repayment	float	Monthly repayment of member	No	No	200
	monthly_gloss_salary	float	Monthly gloss salary of member	No	No	4000
	monthly_net_salary	float	Monthly net salary of member	No	No	3500
	loan_period	date	Period of loan apply	No	No	18/2/2028
	status	ENUM	Status of the application	No	No	approved
	time	date	Submission date of the application	No	No	19/1/2025
	noPF	int	Indicated to the personal profile of member	No	No	4
	member_id (FK 1)	int	Indicated to member ID	No	No	8
	loan_type_id (FK 2)	int	indicated to type of loan	No	No	4

			member apply			
	bank_id (FK 3)	int	Indicated to bank record of member	No	No	6
LoanTypes	loan_type_id (PK)	int	Uniquely identified the type of loan	No	No	5
	loan_type	varchar(255)	Type of loan to be applied	No	No	Simpanan Tetap
Banks	bank_id (PK)	int	Uniquely identified each members' bank info	No	No	8
	bank_name	varchar(255)	Members' bank name	No	No	Maybank
	bank_account	int	Members' bank account number	No	No	123456789000
Individual Report	individualReport_id (PK)	int	Uniquely identified each members' individual report	No	No	8

	member_id (FK)	int	Indicated to which member own the report	No	No	7
	loan_id (FK)	int	Indicated to the loan application that the member made	No	No	2
	savings_id (FK)	int	Indicated to the savings record of the member	No	No	5
Annual Report	annualReport_id (PK)	int	Uniquely identified each annual report uploaded	No	No	4
	filePDF	BLOB	Annual report in PDF file	Yes	No	Binary file

3. Logical Data Model (Logical ERD)

3.1 Logical Data Design (Logical ERD)



3.2 Data Dictionary for logical ERD

3.2.1 Entity: Guest

Attribute Name	Type	Description
guest_id (PK)	int	Uniquely identifies each guest
guestPassword	varchar(255)	Account password for login

guestUsername	varchar(255)	Guests' full name
guestEmail	varchar(255)	Account email for login

3.2.2 Entity: Members

Attribute Name	Type	Description
member_id (PK)	int	Uniquely identifies each member
name	varchar(255)	Members' full name
ic	int	Members' identification card number
email	varchar(255)	Account email for receive notification
sex	varchar(255)	Members' gender
phone	varchar(255)	Members' phone number
DOB	date	Members' birth date
noPF	int	Members' personal profile
address	text	Members' address
status	ENUM	Indicates the application state
city	text	City where the member is located
postcode	int	Postcode where the member is located
state	text	State where the member is located

office_address	text	Office address of member
office_city	text	Office city of member
office_postcode	int	Office postcode of member
time	date	The date of submission application
familyNo	int	Family record of members
staffNo	int	Staff number of member

3.2.3 Entity: Admin

Attribute Name	Type	Description
admin_id (PK)	int	Uniquely identifies each admin
adminPassword	varchar(255)	Account password for admin login
adminUsername	varchar(255)	Admins' full name
adminEmail	varchar(255)	Account email for admin login

3.2.4 Entity: Transaction

Attribute Name	Type	Description
Transaction_id (PK)	int	Uniquely identified transaction record
transactionType	ENUM	Type of transaction made
time	date	Date of transaction made

Savings_id (FK)	int	Indicated to the specific savings account info
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3.2.5 Entity: Savings

Attribute Name	Type	Description
savings_id (PK)	int	Uniquely identified savings record
savingAmount	float	Amount of the member savings
dividendRate	float	Yearly dividen rate
YuranMasuk	float	Amount of members' Yuran Masuk
ModalSyer	float	Amount of members' Modal Syer
ModalYuran	float	Amount of members' Modal Yuran
WangDepositAnggota	float	Amount of members' deposite
Al-Abrar	float	Amount of members' Al-Albrar
SimpananTetap	float	Amount of members' fixed deposit
amount	float	Total amount of the savings

member_id (FK1)	int	Indicated to the member who own the savings record
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3.2.6 Entity: Family

Attribute Name	Type	Description
FamilyNo (PK)	int	Uniquely identified family record of member
name	varchar(255)	Members' family name
IC	varchar(255)	Members' family ic
relationship	varchar(255)	Relationship between the family and member

3.2.7 Entity: Staff Info

Attribute Name	Type	Description
staffNo	int	Uniquely identified each staff
position	varchar(255)	Staffs' position in KADA
gred	varchar(255)	Staffs' grade in KADA

3.2.8 Entity: Guarantor

Attribute Name	Type	Description
GuarantorNo (PK)	int	Uniquely identified guarantor of member
name	varchar(255)	Guarantors' name

ic	varchar(255)	Guarantors' IC
staff_id	int	Guarantors' staff ID in KADA
phone	varchar(255)	Guarantors' phone number

3.2.9 Entity: Loans

Attribute Name	Type	Description
application_id (PK)	int	Uniquely identified each application record
date_apply	date	Submission date of application
loan_amount	float	Amount of loan to apply
interest_rate	float	Interest rate for loan
monthly_repayment	float	Monthly repayment of member
monthly_gloss_salary	float	Monthly gloss salary of member
monthly_net_salary	float	Monthly net salary of member
loan_period	date	Period of loan apply
status	ENUM	Status of the application
time	date	Submission date of the application
noPF	int	Indicated to the personal profile of member

member_id (FK 1)	int	Indicated to member ID
loan_type_id (FK 2)	int	indicated to type of loan member apply
bank_id (FK 3)	int	Indicated to bank record of member

3.2.10 Entity: Loan Types

Attribute Name	Type	Description
loan_type_id (PK)	int	Uniquely identified the type of loan
loan_type	varchar(255)	Type of loan to be applied

3.2.11 Entity: Banks

Attribute Name	Type	Description
bank_id (PK)	int	Uniquely identified each members' bank info
bank_name	varchar(255)	Members' bank name
bank_account	int	Members' bank account number

3.2.12 Entity: Individual Report

Attribute Name	Type	Description
individualReport_id (PK)	int	Uniquely identified each members' individual report
member_id (FK)	int	Indicated to which member own the report

loan_id (FK)	int	Indicated to the loan application that the member made
savings_id (FK)	int	Indicated to the savings record of the member

3.2.13 Entity: Annual Report

Attribute Name	Type	Description
annualReport_id (PK)	int	Uniquely identified each annual report uploaded
filePDF	BLOB	Annual report in PDF file

3.3 Relational Database Schemas

1. Guest (guest_id, guestPassword, guestUsername, guestEmail)

Primary Key: guest_id

2. Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city, postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

Primary Key: member_id

Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

3. Admin (admin_id, adminPassword, adminUsername, adminEmail)

Primary Key: admin_id

4. Family (familyNo, name, IC, relationship)

Primary Key: familyNo

5. StaffInfo (staffNo, position, gred)
Primary Key: staffNo
6. Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)
Primary Key: savings_id
Foreign Key: member_id reference Member(member_id)
7. Transaction(transaction_id, transactionType, time, savings_id, loan_id)
Primary Key: transaction_id
Foreign Key: loan_id reference Loan(loan_id), savings_id reference Savings(savings_id)
8. Guarantor (GuarantorNo, name, ic, staff_id, phone)
Primary Key: GuarantorNo
9. Loan(loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id)
Primary Key: loan_id
Foreign Key: member_id reference Member(member_id), GuarantorNo reference Guarantor(GuarantorNo), loan_type_id reference Loan Types(loan_type_id), bank_id reference Bank(bank_id)
10. LoanTypes (loan_type_id, loan_type)
Primary Key: loan_type_id
11. Banks (bank_id, bank_name, bank_account)
Primary Key: bank_id
12. IndividualReport (individualReport_id, member_id, loan_id, savings_id)

Primary Key: individualReport_id

Foreign Key: member_id reference Member(member_id), loan_id reference

Loan(loan_id), savings_id reference Savings(savings_id)

13. AnnualReport (annualReport_id, filePDF)

Primary Key: annualReport_id

3.4 Normalization

Member Table

1NF

MemberForm (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id, familyNo, name, IC, relationship, staffNo, position, gred, savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount)

Primary Key: member_id

2NF

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id, familyNo, name, IC, relationship, staffNo, position, gred)

Primary Key: member_id

Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)

Primary Key: savings_id

Foreign Key: member_id reference Member(member_id)

3NF

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

Primary Key: member_id

Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

Savings (savings_id, savingAmount, dividendRate, YuranMasuk, ModalSyer, ModalYuran, Wang DepositAnggota, Al-Abrar, SimpananTetap, amount, member_id)

Primary Key: savings_id

Foreign Key: member_id reference Member(member_id)

Family (familyNo, name, IC, relationship)

Primary Key: familyNo

Staff Info (staffNo, position, gred)

Primary Key: staffNo

Admin Table

1NF

Admin (admin_id, adminPassword, adminUsername, adminEmail)

Primary Key: admin_id

2NF

All the derived relations are already in Second Normal Form(2NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as First Normal Form (1NF).

3NF

All the derived relations are already in Third Normal Form(3NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as Second Normal Form (2NF).

Guest Table

1NF

Guest (guest_id, guestPassword, guestUsername, guestEmail)

Primary Key: guest_id

2NF

All the derived relations are already in Second Normal Form(2NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as First Normal Form (1NF).

3NF

All the derived relations are already in Third Normal Form(3NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as Second Normal Form (2NF).

Loan table

1NF

Loans (loan_id, member_id, loan_type, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_name, guarantor_ic, guarantor_phone,guarantor_staffNo)

Primary Key: loan_id

Foreign Key: member_id references Member (member_id)

2NF

Loans (loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_name, guarantor_ic, guarantor_phone,guarantor_staffNo)

Primary Key: loan_id

Foreign Keys:member_id references Member (member_id),loan_type_id references

Loan_Types(loan_type_id),guarantor_no references Guarantors(guarantor_no)

Loan_Types (loan_type_id, loan_type)

Primary Key: loan_type_id

3NF

Loans (loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_no)

Primary Key: loan_id

Foreign Keys:member_id references Member (member_id),loan_type_id references

Loan_Types(loan_type_id),guarantor_no references Guarantors(guarantor_no)

Loan_Types (loan_type_id, loan_type)

Primary Key: loan_type_id

Guarantors (guarantor_no, name, ic, phone, staffID)

Primary Key: guarantor_id

Transaction table

1NF

Transactions (Transaction_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, guest_id, family_name, family_IC, relationship, position, gred, Transaction_type, loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time, noP , amount, time, loan_type, guarantor_name, guarantor_ic, guarantor_phone,guarantor_staffNo)

Primary Key: Transaction_id

2NF

Transactions (Transaction_id, member_id, Transaction_type, loan_id, amount, time)

Primary Key: Transaction_id

Foreign Keys: member_id references Member (member_id), loan_id references

Loans(loan_id)

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city, postcode, state, office_address, office_city, office_postcode, time, staffNo, guest_id, family_name, family_IC, relationship, position, gred)

Primary Key: member_id

Loans (loan_id, member_id, loan_type, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time, noPF, guarantor_name, guarantor_ic, guarantor_phone, guarantor_staffNo)

Primary Key: loan_id

Foreign Key: member_id references Member (member_id)

3NF

Transactions (Transaction_id, member_id, Transaction_type, loan_id, amount, time)

Primary Key: Transaction_id

Foreign Keys: member_id references Member (member_id), loan_id references

Loans(loan_id)

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city, postcode, state, office_address, office_city, office_postcode, time, guest_id, staffNo, familyNo)

Primary Key: member_id

Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

Family (familyNo, name, IC, relationship)

Primary Key: familyNo

StaffInfo (staffNo, position, gred)

Primary Key: staffNo

Loans (loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_no)

Primary Key: loan_id

Foreign Keys:member_id references Member (member_id),loan_type_id references

Loan_Types(loan_type_id),guarantor_no references Guarantors(guarantor_no)

Loan_Types (loan_type_id, loan_type)

Primary Key: loan_type_id

Guarantors (guarantor_no, name, ic, phone, staffID)

Primary Key: guarantor_id

Bank Table

1NF

Banks (bank_id, bank_name, bank_account)

Primary Key: bank_id

2NF

All the derived relations are already in Second Normal Form(2NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies

will stay unchanged which same as First Normal Form (1NF).

3NF

All the derived relations are already in Third Normal Form(3NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as Second Normal Form (2NF).

Individual Report Table

1NF

IndividualReport (individualReport_id, member_id, loan_id, savings_id)

Primary Key: individualReport_id

Foreign Key: member_id reference Member(member_id), loan_id reference Loan(loan_id), savings_id reference Savings(savings_id)

2NF

All the derived relations are already in Second Normal Form(2NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as First Normal Form (1NF).

3NF

All the derived relations are already in Third Normal Form(3NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as Second Normal Form (2NF).

Annual Report Table

1NF

AnnualReport (annualReport_id, filePDF)

Primary Key: annualReport_id

2NF

All the derived relations are already in Second Normal Form(2NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as First Normal Form (1NF).

3NF

All the derived relations are already in Third Normal Form(3NF) as there is no partial dependencies exist in the relation. Therefore, all the derived dependencies will stay unchanged which same as Second Normal Form (2NF).

4.Interface design and SQL implementation

4.1 Interface Design

The Interface Design has been divided into 5 different modules done by each member, which is Member Management Module, Loan Management Module, Savings and Dividend Management Module, Reporting Module and Administrative Module.

4.1.1 Member Management Module

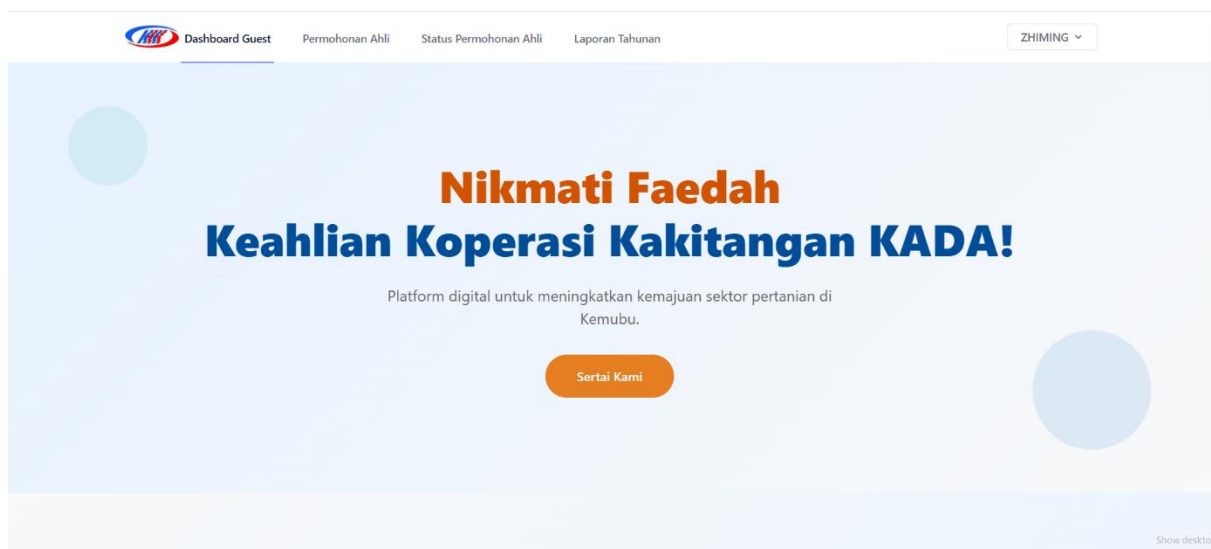


Figure 4.1.1.1: Guest Dashboard Page

Maklumat Peribadi

Nama Pendaftar: No. Anggota:

No. K/P: No. Telefon:

Email:

Alamat:

Bandar: Poskod: Negeri:

[Seterusnya >](#)

Figure 4.1.1.2: Member Registration Form (Section Maklumat Peribadi)

Maklumat Pekerjaan & Peribadi

Jantina: Tarikh Lahir:

Agama: Bangsa:

Jawatan: Gred:

No. Fail Peribadi: Gaji (RM):

Alamat Pejabat:

Bandar: Poskod: Negeri:

Figure 4.1.1.3: Member Registration Form (Section Maklumat Pekerjaan & Peribadi)

The screenshot shows the 'Maklumat Keluarga' (Family Information) section of a registration form. It features a table with columns: BIL (No.), NAMA (Name), NO. K/P (K/P Number), HUBUNGAN (Relationship), and TINDAKAN (Action). The first row is for a family member with the name 'Nama ahli ke', K/P number 'No. K/P', and relationship 'Pilih Hu'. Below the table is a '+ Tambah Ahli Keluarga' (Add Family Member) button. Navigation buttons 'Sebelumnya' (Previous) and 'Seterusnya' (Next) are at the bottom.

BIL	NAMA	NO. K/P	HUBUNGAN	TINDAKAN
1	Nama ahli ke Sila masukkan nama	No. K/P Sila masukkan no. kad pengenalan	Pilih Hu Sila pilih hubungan	

+ Tambah Ahli Keluarga

< Sebelumnya Seterusnya >

Figure 4.1.1.4: Member Registration Form (Section Maklumat Keluarga)

The screenshot shows the 'Yuran dan Sumbangan' (Fees and Contributions) section. It contains a table with columns: BIL (No.), PERIODE (Period), and NILAI (Value). The first row is for 'Yuran Biasa' (Regular Fee) with a value of 50. Below the table is a 'Kandah Pembayaran' (Payment Method) section with a dropdown menu set to 'Bank' and a 'Pembayaran Dalam Talian' (Online Payment) button. Navigation buttons 'Sebelumnya' (Previous) and 'Seterusnya' (Next) are at the bottom.

BIL	PERIODE	NILAI
1	Yuran Biasa	50
2	Maka Nya (Sila masukkan)	100
3	Maka Nya (Sila masukkan)	20
4	Maka Nya (Sila masukkan)	20
5	Sumbangan (Sila masukkan)	5

Kandah Pembayaran: Bank Pembayaran Dalam Talian

< Sebelunya Seterusnya >

Figure 4.1.1.5: Member Registration Form (Section Yuran dan Sumbangan)

The screenshot shows the 'STATUS PERMOHONAN AHLI' (Member Application Status) page. It features a progress bar with three stages: 'Permohonan Dihantar' (Application Submitted), 'Telah Diproses' (Already Processed), and 'Ditolak' (Rejected). Below the progress bar is a form for 'Maklumat Peribadi' (Personal Information) with fields for Nama (Name), No. K/P (K/P Number), Tarikh Lahir (Date of Birth), Jantina (Gender), Agama (Religion), Bangsa (Race), No. Telefon (Phone Number), and Email.

STATUS PERMOHONAN AHLI

Permohonan Dihantar Telah Diproses Ditolak

Maklumat Peribadi

Nama: Sabrina No. K/P: 041118011234

Tarikh Lahir: 2004-11-18 Jantina: Lelaki Agama: Kristian Bangsa: India

No. Telefon: Email:

Figure 4.1.1.6: Status Member Registration

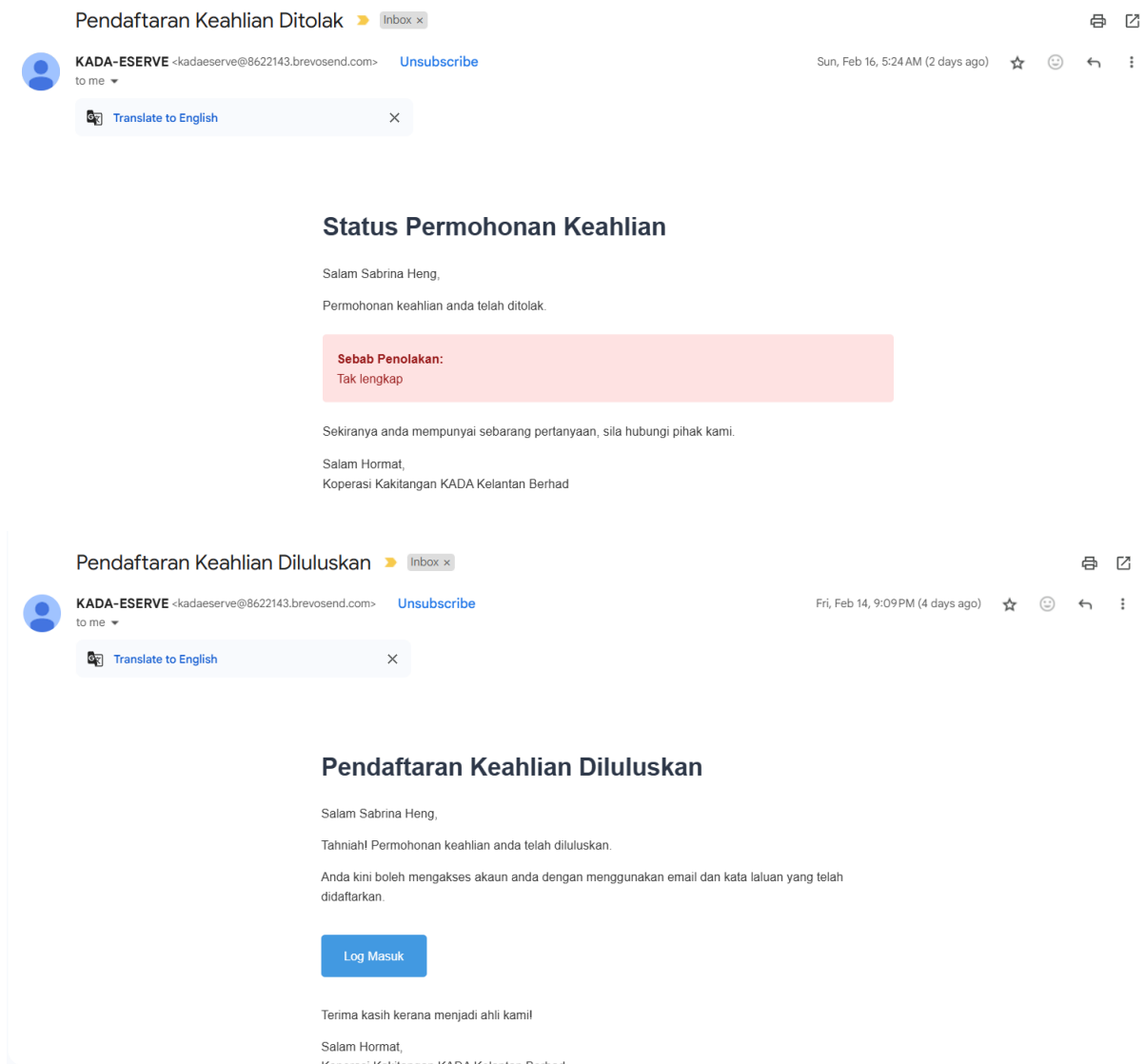


Figure 4.1.1.7: Email about Approve/Reject Member Registration

4.1.2 Loan Management Module

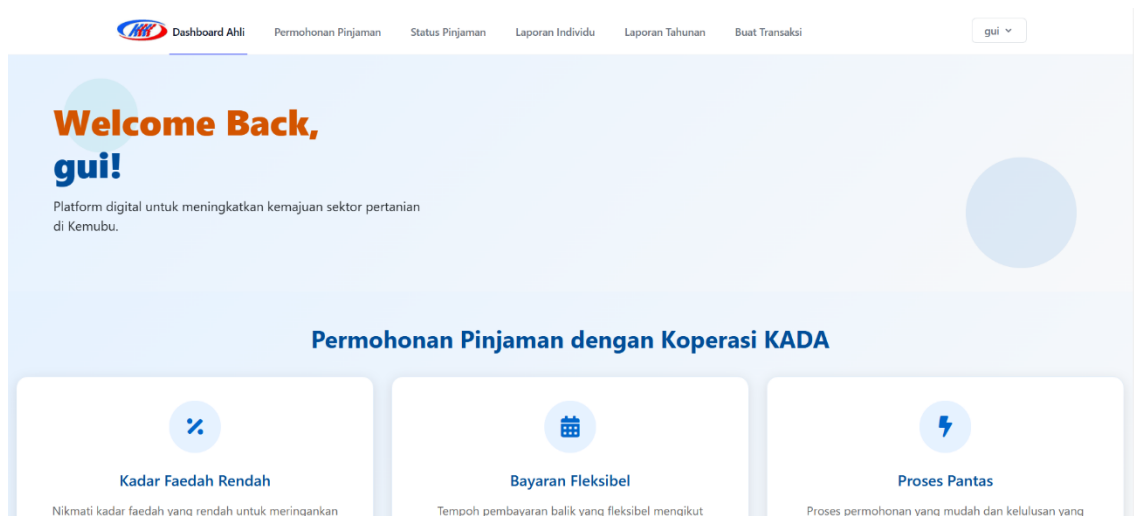


Figure 4.1.2.1: Member Dashboard Page

The screenshot shows the 'Permohonan Pinjaman' (Loan Application) form, specifically the 'Maklumat Peribadi' (Personal Information) section. The form is structured with a progress indicator at the top showing four steps: 1. Maklumat Peribadi (active), 2. Maklumat Pinjaman, 3. Maklumat Penjamin, and 4. Pengesahan. The 'Maklumat Peribadi' section contains several input fields: 'Nama' (Name) with the value 'gui', 'No. Kad Pengenalan' (IC Number) with the value '040823040326', 'Alamat' (Address) with the value 'Tangkak Johor', 'No. Telefon' (Phone Number) with the value '0129715408', and 'Emel' (Email) with the value 'guisin@graduate.utm.my'. A blue button labeled 'Seterusnya >' (Next) is located at the bottom right of the form.

Figure 4.1.2.2: Loan Application Form (Section Maklumat Peribadi)

1

2

3

4

Maklumat PeribadiMaklumat PinjamanMaklumat PenjaminPengesahan

Maklumat Pinjaman

Jenis Pembiayaan

Pilih Jenis Pembiayaan

Bank

Pilih Bank

No. Akaun Bank

Contoh: 1234567890

Jumlah Pinjaman (RM)

RM

Tempoh Pinjaman (Bulan)

Gaji Kasar Bulanan (RM)

RM

Gaji Bersih Bulanan (RM)

RM

< Sebelumnya

Seterusnya >

Figure 4.1.2.3: Loan Application Form (Section Maklumat Pinjaman)

1

2

3

4

Maklumat PeribadiMaklumat PinjamanMaklumat PenjaminPengesahan

Maklumat Penjamin

Penjamin Pertama

Nama

gui2

No. PF

112

No. Kad Pengenalan

040823040324

No. Telefon

0129715408

No. Anggota

1235

Penjamin Kedua

Nama

gui3

No. PF

113

No. Kad Pengenalan

040823040328

No. Telefon

0129715408

No. Anggota

1236

< Kembali

Seterusnya >

Figure 4.1.2.4: Loan Application Form (Section Maklumat Penjamin)

1

2

3

4

Maklumat Peribadi
Maklumat Pinjaman
Maklumat Penjamin
Pengesahan

Pengesahan

Maklumat Peribadi

Nama

gui

No. Kad Pengenalan

040823040326

Alamat

Tangkak Johor

No. Telefon

0129715408

Email

gui@graduate.utm.my

Maklumat Pinjaman

Jenis Pembayaran	Jumlah Pinjaman
Skim Khas	RM 1000
Tempoh Pinjaman	Bank
12 bulan	RHB Bank Berhad
No. Akaun Bank	Gaji Kasar Bulanan
12345678	RM 8000
Gaji Bersih Bulanan	
RM 7500	

Maklumat Penjamin

Maklumat Penjamin

Penjamin Pertama

Nama

gui2

No. Kad Pengenalan

040823040324

No. Anggota

1235

No. PF

112

No. Telefon

0129715408

Penjamin Kedua

Nama

gui3

No. Kad Pengenalan

040823040328

No. Anggota

1236

No. PF

113

No. Telefon

0129715408

☒ Saya mengesahkan bahawa semua maklumat yang diberikan adalah benar dan tepat

<
Sebelumnya

Hanitar Permohonan

Figure 4.1.2.5: Loan Application Form (Section Pengesahan)

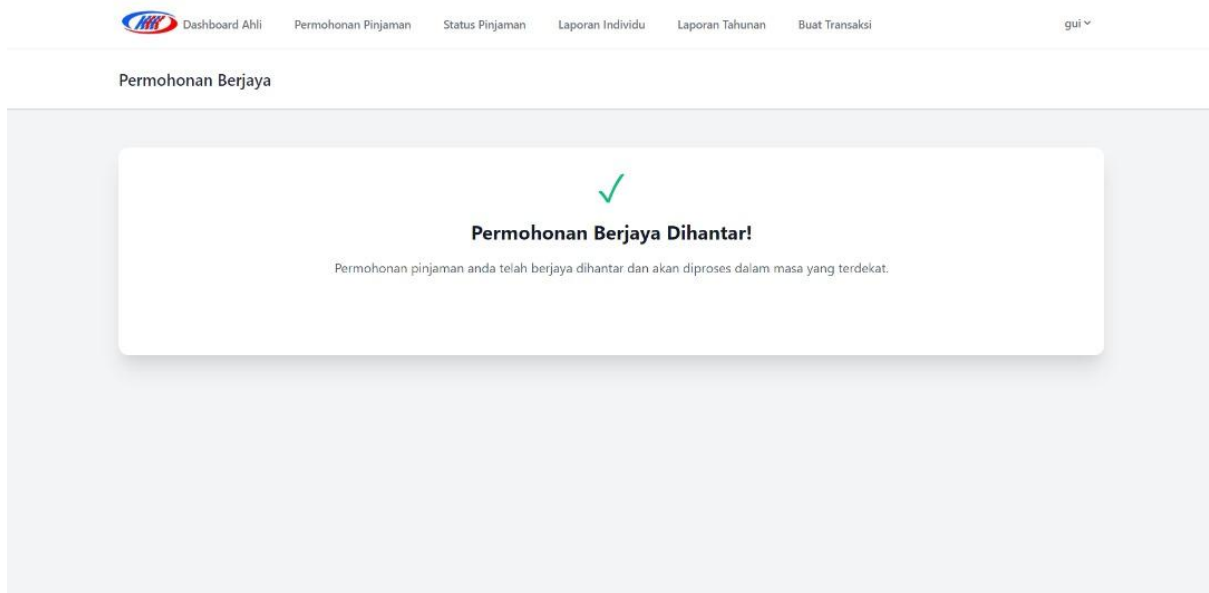


Figure 4.1.2.6: Loan Application Form Successfully Submitted

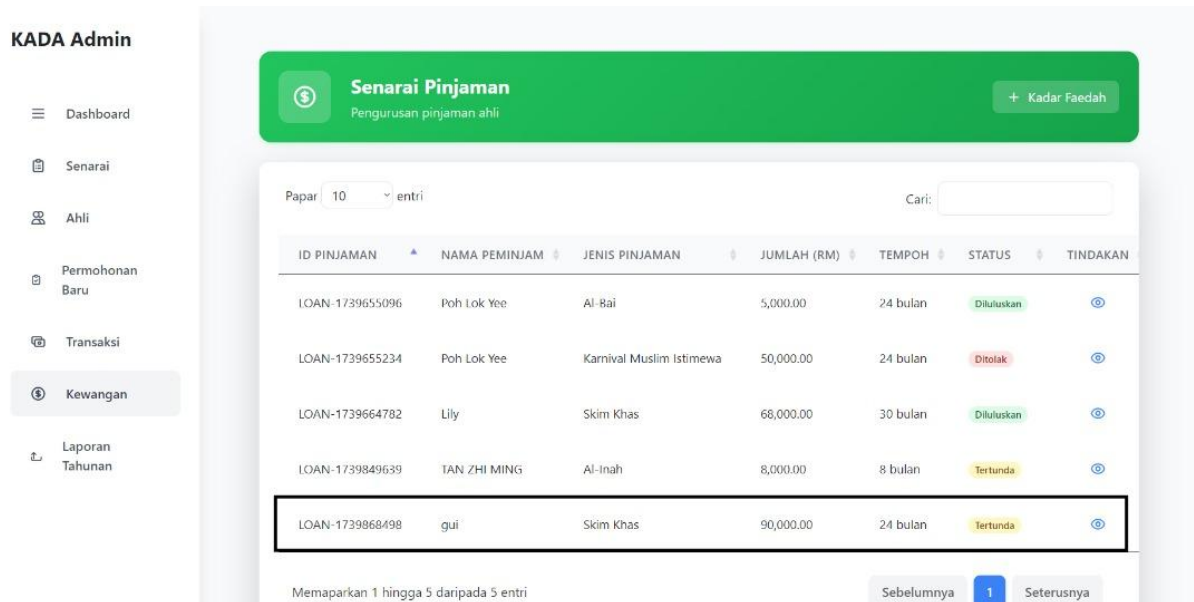


Figure 4.1.2.7: Member Loan Application List (Admin)

KADA Admin

Dashboard

Senarai

Ahli

Permohonan Baru

Transaksi

Kewangan

Laporan Tahunan

Kembali ke Senarai Pinjaman

Lulus Pinjaman

Tolak Pinjaman

Maklumat Pinjaman

ID Pinjaman

LOAN-1739868498

Status Pinjaman

Tertunda

Jenis Pinjaman

Skim Khas

Tarikh Mohon

2025-02-18 00:00:00

Jumlah Pinjaman

RM 90,000.00

Tempoh Pinjaman

24 bulan

Kadar Faedah

5%

Bayaran Bulanan

RM 3,948.43

Jumlah Bayaran

RM 94,762.32

Maklumat Peribadi

Nama Penuh

gui

No. KP

040823040326

Email

guisin@graduate.utm.my

No. Telefon

0129715408

Alamat Rumah

Tangkak Johor

Bandar

Tangkak

Poskod

84900

Negeri

Johor

Maklumat Pekerjaan

Alamat Pejabat

Kelantan

Bandar

Kelantan

Poskod

81000

Gaji Kasar Bulanan

RM 9,000.00

Gaji Bersih Bulanan

RM 8,500.00

Maklumat Bank

Nama Bank

Affin Islamic Bank Berhad

No. Akaun Bank

5044299236

Maklumat Penjamin

Penjamin 1

Nama Penuh

Gui3

No. Kad Pengenalan

780920015110

No. PF

113

No. Telefon

0182222222

No. Anggota

001

Penjamin 2

Nama Penuh

Gui2

No. Kad Pengenalan

770920015010

No. PF

112

No. Telefon

0172222222

No. Anggota

999

Figure 4.1.2.8: Member Loan Application Details and Action Buttons (Admin)

44

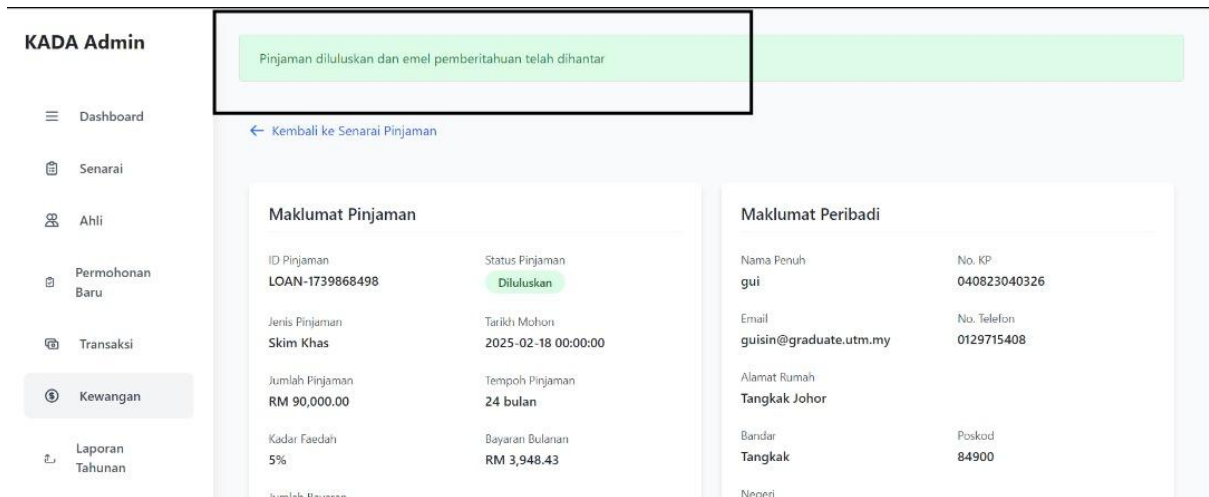


Figure 4.1.2.9: Success Message after Approve Loan Application (Admin)

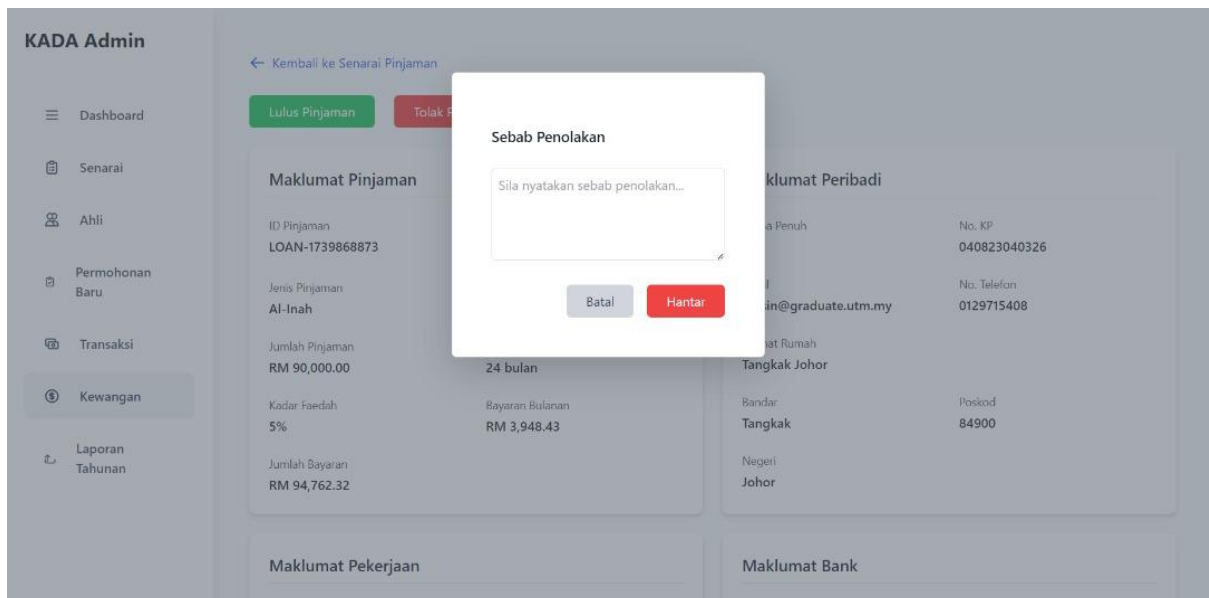


Figure 4.1.2.10: Rejected Reason Box (Admin)

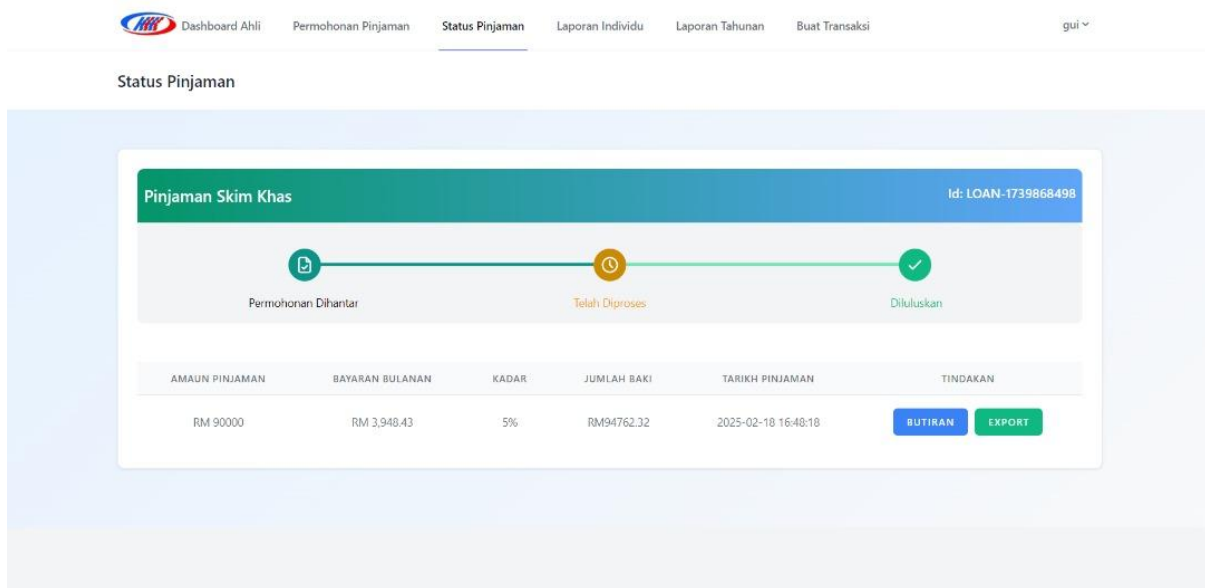


Figure 4.1.2.11: Loan Application Status

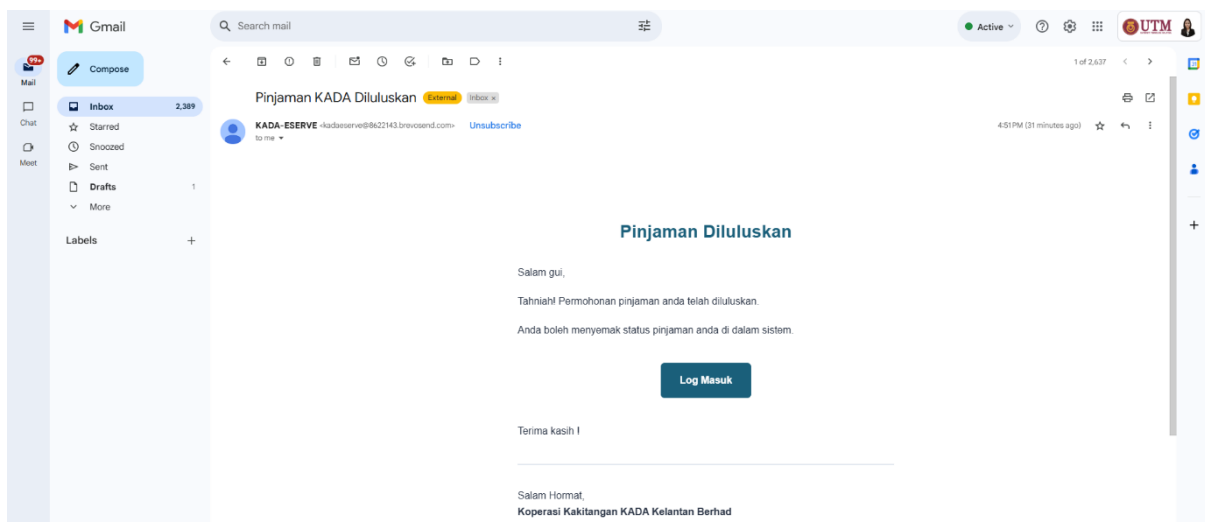


Figure 4.1.2.12: Email about Approve/Reject Loan Application


Dashboard Ahli
Permohonan Pinjaman
Status Pinjaman
Laporan Individu
Laporan Tahunan
Buat Transaksi
gui

Buat Transaksi Baru

Maklumat Transaksi

Sila lengkapkan butiran transaksi anda

Jenis Transaksi

Bayaran Pinjaman

Pilih Pinjaman

Pinjaman #LOAN-1739868498 - Baki: RM94762.32

Jumlah (RM)

100

Kaedah Pembayaran

☐ Online Banking
☒ Tunai

Hantar Transaksi

Figure 4.1.2.13: Transaction for Loan Application

KADA Admin

- Dashboard
- Senarai
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan
- Laporan Tahunan

Senarai Transaksi

Pengurusan transaksi ahli

Papar 10 entri

Cari:

ID TRANSAKSI	NAMA AHLI	JENIS	JUMLAH (RM)	STATUS	TARIKH	TINDAKAN
TRX202502160533582865	Poh Lok Yee	Bayaran Pinjaman	500.00	Diluluskan	16/02/2025 05:33	
TRX202502160537487880	Poh Lok Yee	Bayaran Pinjaman	4,000.00	Ditolak	16/02/2025 05:37	
TRX202502160804554183	Poh Lok Yee	Tabung Anggota	200.00	Tertunda	16/02/2025 08:04	
TRX202502160809123810	Poh Lok Yee	Bayaran Pinjaman	150.00	Tertunda	16/02/2025 08:09	
TRX202502160809482682	Poh Lok Yee	Bayaran Pinjaman	150.00	Diluluskan	16/02/2025 08:09	
TRX202502161122031767	Poh Lok Yee	Bayaran Pinjaman	1,000.00	Tertunda	16/02/2025 11:22	
TRX202502181720596932	gui	Bayaran Pinjaman	100.00	Tertunda	18/02/2025 17:20	

Figure 4.1.2.14: Transaction List (Admin)

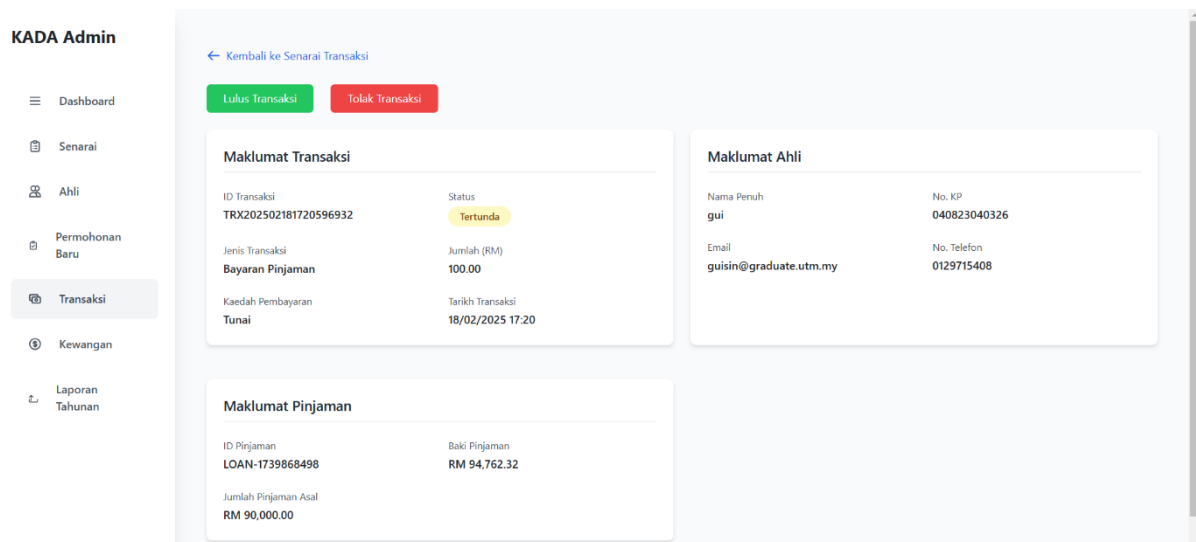


Figure 4.1.2.15: Approve/Reject Transaction for Loan Application (Admin)

4.1.3 Savings and Dividend Management Module

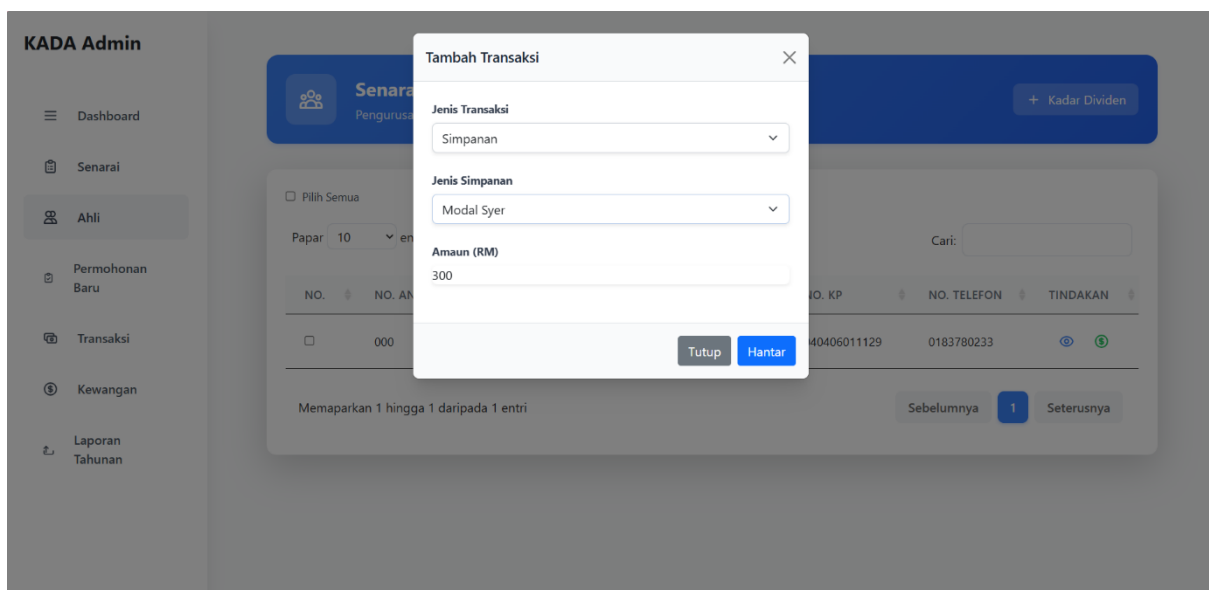


Figure 4.1.3.1: Admin update a member's transaction

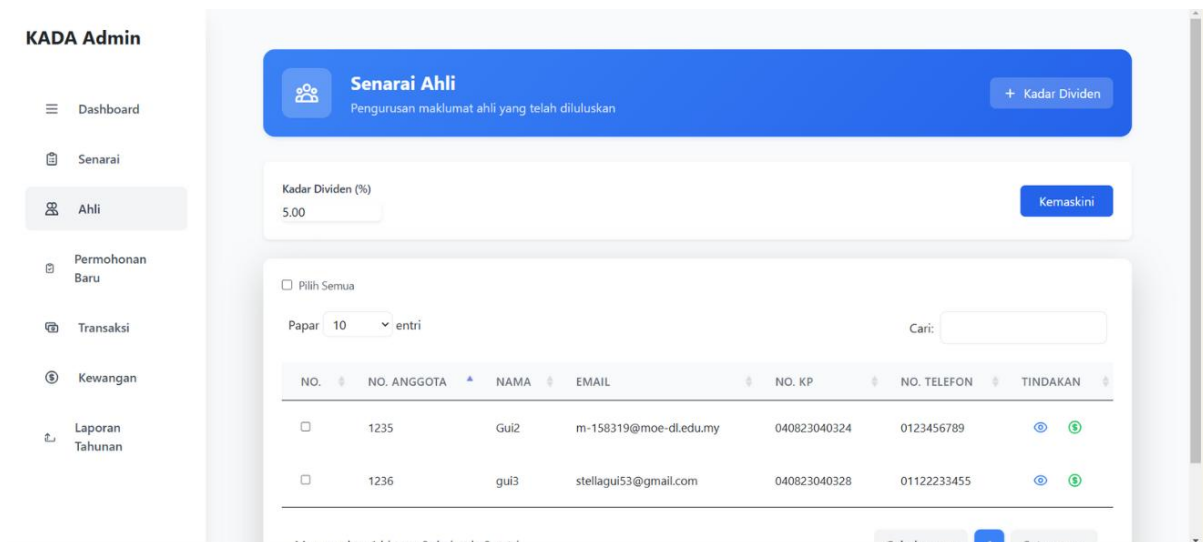


Figure 4.1.3.2: Admin update dividend rate for members

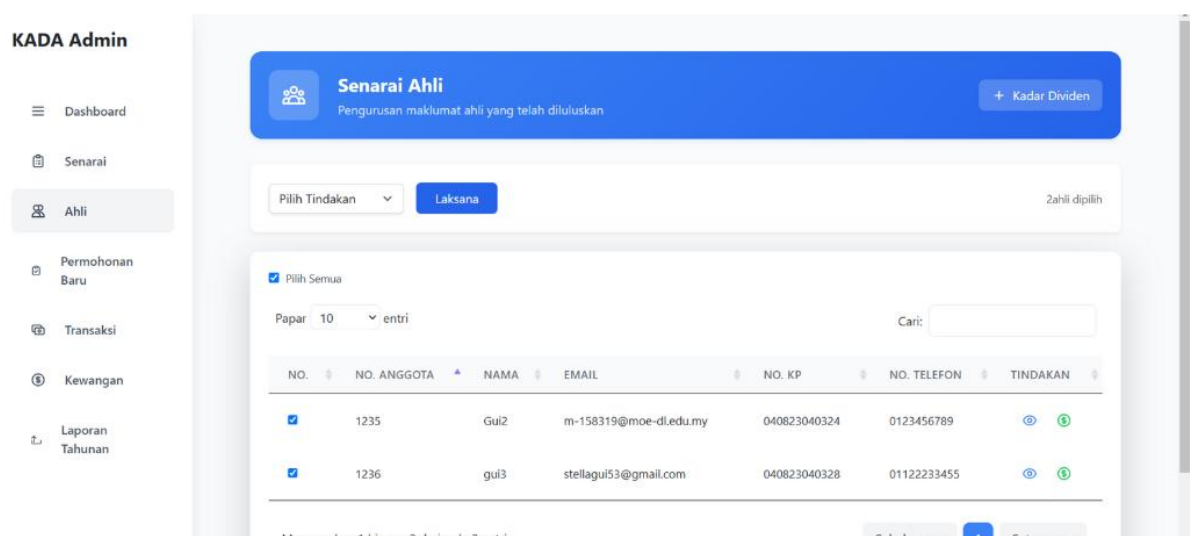


Figure 4.1.3.3: Filter members for intended action of updating rate

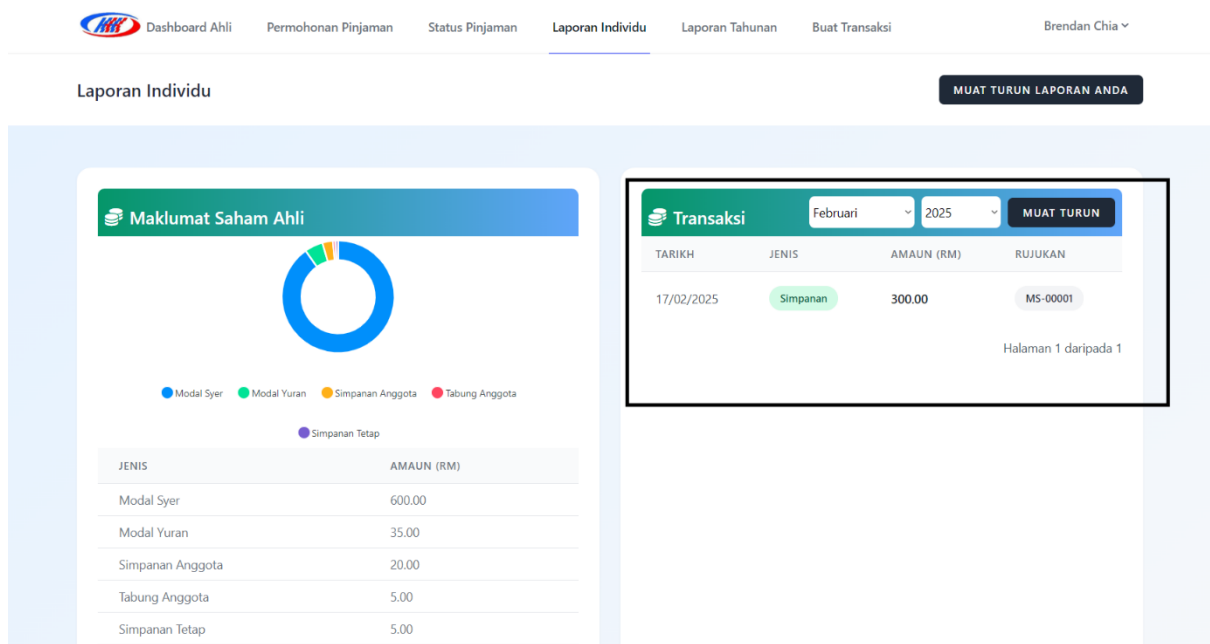


Figure 4.1.3.4: Display the transactions made by a member

4.1.4 Reporting Module

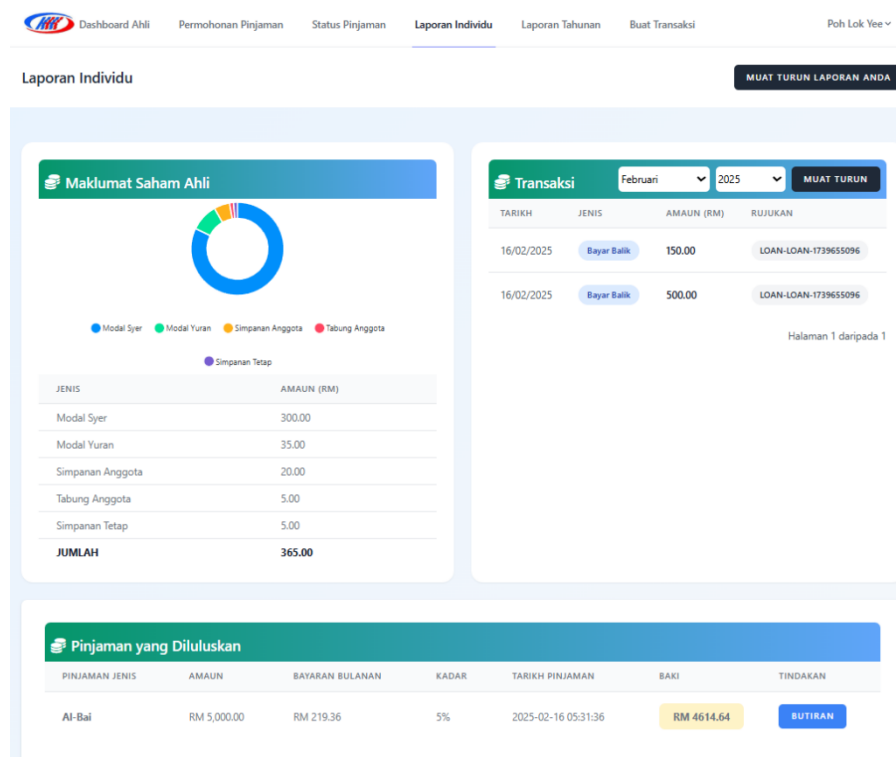


Figure 4.1.4.1: Individual Report Page Interface which consist the Savings, Transaction and Approved Loans details of Member



Nama: **Leong Foong Har**

IC: **111111111111** | No Ahli: **1234**

Tuan/Puan,

**PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA
KELANTAN BERHAD BAGI TAHUN BERAKHIR 28-02-2025**

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah
sebagaimana berikut:

MAKLUMAT SAHAM AHLI

Modal Syer	Modal Yuran	Simpanan Tetap	Tabung Kebajikan	Simpanan Anggota
42.00	2.00	42.00	42.00	12.00

MAKLUMAT PINJAMAN AHLI

Pinjaman Jenis	Amaun	Bayaran Bulanan	Kadar
TIADA REKOD PINJAMAN			

**PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA
KELANTAN BERHAD BAGI TAHUN BERAKHIR 28-02-2025**

Saya, **Leong Foong Har**, No. Ahli: **1234**, mengesahkan bahawa Penyata Kewangan
Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir **28-02-2025** adalah seperti
di atas.

Tarikh: **10-02-2025**

Nota: Sekiranya pihak Koperasi tidak menerima maklum balas daripada tuan/puan sehingga **28-02-2025**, penyata ini
dianggap diterima.

Tuan/Puan dapat menghubungi pejabat Koperasi di talian 09-748 2000 untuk sebarang pertanyaan. Terima kasih.

Figure 4.1.4.2: Individual Report Export in File


Dashboard Ahli
Permohonan Pinjaman
Status Pinjaman
Laporan Individu
Laporan Tahunan
Buat Transaksi
Poh Lok Yee

Pinjaman **Al-Bai**
DILULUSKAN
KEMBALI

Maklumat Pinjaman
Baki Pinjaman / Jumlah Bayaran Balik: RM 4,614.64 / RM 5,264.64

Pinjaman ID	:LOAN-1739655096	Gaji Bersih	:RM 4,000.00
Amaun Pinjaman	:RM 5,000.00	Gaji Kasar	:RM 4,500.00
Bayaran Bulanan	:RM 219.36	Kadar	:5%
Tempoh	:24 bulan	Tarikh Pinjaman	:2025-02-16 05:31:36

Maklumat Bank

Nama Bank	: Affin Islamic Bank Berhad	Akaun No	: 4125415
-----------	-----------------------------	----------	-----------

Maklumat Penjamin

Penjamin Pertama		Penjamin Kedua	
Nama	: Sabrina Heng Wei Qi	Nama	: Tan Zhi Ming
IC	: 111111111111	IC	: 253698745112
No telefon	: 01111111111	No Telefon	: 01164287954
No Anggota	: A001	No Anggota	: 1928
No PF	: A001	No PF	: 1928

Figure 4.1.4.3: Interface for Each Approved Loan Details

Profil

Maklumat Peribadi

AHLI

Nama

Poh Lok Yee

IC

040923140332

Tarikh Lahir

2004-09-23

Agama

Buddha

Bangsa

Cina

No PF

0000

Jawatan

Pekerja

Gred

A

Telefon

01111510260

Email

lokyee1418@gmail.com

Kemas Kini Maklumat

Sila kemas kini maklumat terkini anda. Selepas menekan butang kemaskini, anda dapat mengubah maklumat anda. Jangan lupa menekan butang Simpan.

Alamat Rumah

C-5-8,Pangsapuri Bayu,Jalan Mawar,Taman Bukit Serdang

Bandar

SERI KEMBANGAN

Poskod

43300

Negeri

Selangor

Alamat Pejabat

C-5-8,Pangsapuri Bayu,Jalan Mawar,Taman Bukit Serdang

Bandar Pejabat

SERI KEMBANGAN

Poskod Pejabat

43300

Negeri Pejabat

Selangor

KEMAS KINI

Figure 4.1.4.4: Profile Page That Display Member's Data

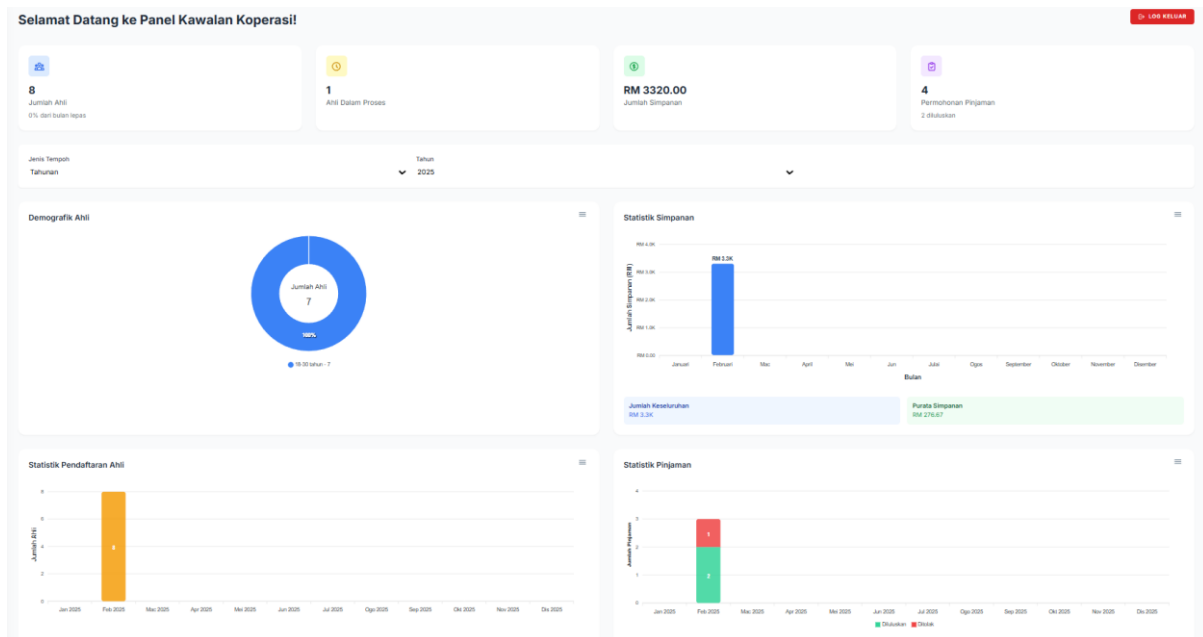


Figure 4.1.4.5: Admin Dashboard That Display the Data Trends and Summary of The System

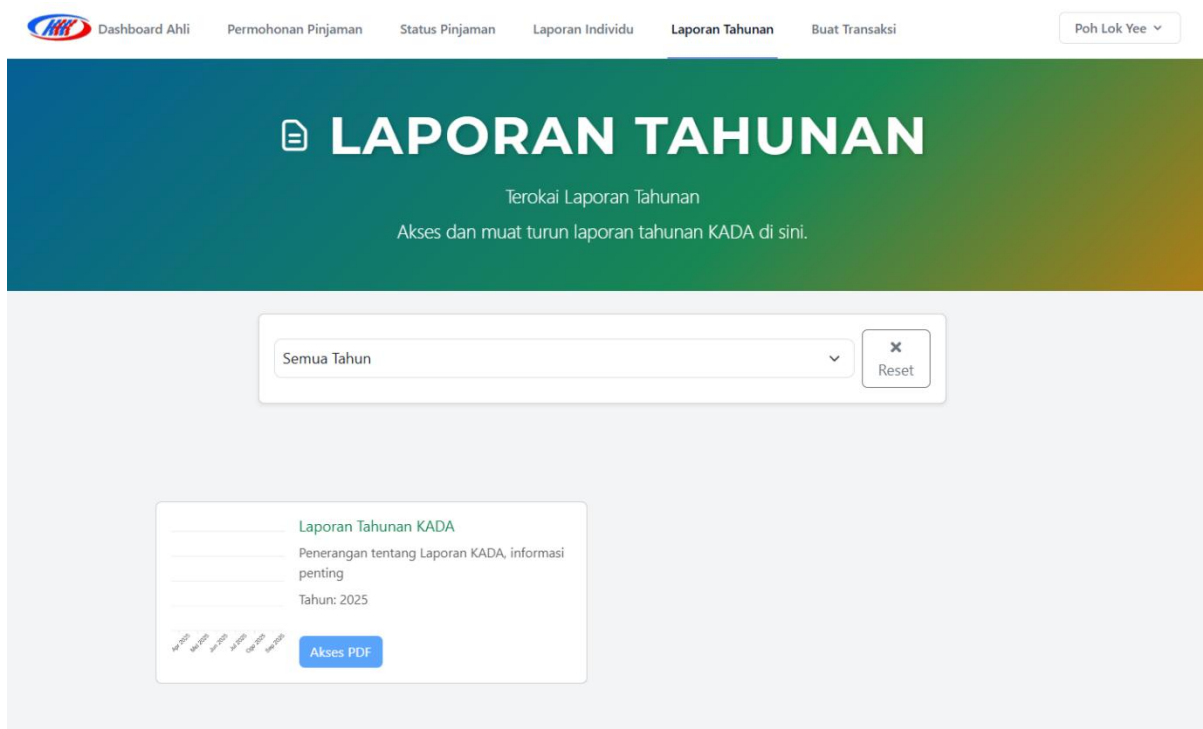


Figure 4.1.4.6: Annual Report Display Page

KADA Admin

Dashboard
Senarai
Ahli
Permohonan Baru
Transaksi
Kewangan
Laporan Tahunan

Laporan Tahunan
Pengurusan laporan tahunan

Paparan Laporan

MUAT NAIK FAIL

Tajuk

Penerangan

Tahun

2025

Pilih imej Muka Depan PNG, PNG sehingga 5MB

Pilih Fail PDF PDF sehingga 100MB

Muat Naik

LAPORAN DIMUAT NAIK

1 laporan dijumpai

TAHUN	TAJUK	TINDAKAN
2025	Laporan Tahunan KADA	Kemaskini Padam

Figure 4.1.4.7: Annual Report Upload Section in Admin Page

Kemaskini Laporan Tahunan
Sila ubah maklumat laporan tahunan mengikut keperluan

KEMASKINI FAIL

Tajuk

Laporan Tahunan KADA

Penerangan

Penerangan tentang Laporan KADA, informasi penting

Tahun

2025

Thumbnail (Imej)

[Lihat Thumbnail Semasa](#)

Pilih Fail
Tiada fail dipilih

Biarkan kosong jika tidak mahu menukar thumbnail

Fail Laporan (PDF)

[Lihat PDF Semasa](#)

Pilih Fail
Tiada fail dipilih

Biarkan kosong jika tidak mahu menukar fail PDF

Kemaskini Laporan

Batal

Figure 4.1.4.8: Annual Report Edit Section

55

4.1.5 Administrative Module

KADA Admin

- Dashboard
- Senarai**
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan
- Laporan Tahunan

Senarai Pemohon

Pengurusan status pemohon

Status: Semua Status

Papar: 10 entri

Cari:

NO. ANGGOTA	NAMA	EMAIL	NO. TELEFON	STATUS	TINDAKAN
0000	Poh Lok Yee	lokyee1418@gmail.com	01111510260	Ahli	Lihat Edit
1234	gui	guisin@graduate.utm.my	0129715408	Ahli	Lihat Edit
1235	Gui2	m-158319@moe-dl.edu.my	0129715408	Ahli	Lihat Edit
1236	Gui3	stellagui53@gmail.com	0129715408	Ahli	Lihat Edit
9999	Leong Foong Har	poh.yee@graduate.utm.my	0122848992	Sedang Diproses	Lihat Edit
A001	Sabrina Heng	sabrinaheng115@gmail.com	0126013166	Ahli	Lihat Edit

Figure 4.1.5.1: Resignation Management Page

KADA Admin

- Dashboard
- Senarai**
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan
- Laporan Tahunan

[← Kembali ke Senarai Pemohon](#)

Tiada permohonan berhenti yang...

Sejarah Permohonan Berhenti

NO. AHLI	NAMA	PERMOHONAN	STATUS	TINDAKAN
A23CS0189	TAN ZH	2025 09-27	Diluluskan	Lihat Sebab

Senarai Sebab Berhenti

Sebab 1:
xx

Sebab 2:
xx

Sebab 3:
xx

Tutup

Figure 4.1.5.2: Member Resignation Application Detail

KADA Admin

Dashboard

Senarai

Ahli

Permohonan Baru

Transaksi

Kewangan

Laporan Tahunan



Pendaftaran Menunggu

Senarai pendaftaran yang menunggu kelulusan

+ Tambah Ahli Baru

Papar 10 entri

Cari:

NAMA	NO. KP	EMAIL	NO. TELEFON	TARIKH MOHON	TINDAKAN
Leong Foong Har	090305256881	poh.yee@graduate.utm.my	0122848992	2025-02-16 08:19:52	👁

Memaparkan 1 hingga 1 daripada 1 entri

Sebelumnya

1

Seterusnya

Figure 4.1.5.3: Member Registration Application Management

KADA Admin

Dashboard

Senarai

Ahli

Permohonan Baru

Transaksi

Kewangan

Laporan Tahunan

[← Kembali ke Pendaftaran Menunggu](#)

Lulus Pendaftaran

Tolak Pendaftaran

Maklumat Peribadi

No. Ahli
9999

Nama
Leong Foong Har

No. KP
090305256881

Email
poh.yee@graduate.utm.my

No. Telefon
0122848992

Jantina
Perempuan

Tarikh Lahir
2025-02-12

Agama
Buddha

Bangsa

Cina

Alamat Rumah

C-5-8,Pangsapuri Bayu,Jalan Mawar,Taman Bukit Serdang
43300 SERI KEMBANGAN
Selangor

Maklumat Pekerjaan

Jawatan
Pekerja

Gred
A

No. PF
9999

Gaji
RM 4,500.00

Alamat Pejabat

C-5-8,Pangsapuri Bayu,Jalan Mawar,Taman Bukit Serdang
43300 SERI KEMBANGAN
Selangor

Figure 4.1.5.4: Member Registration Application Detail

Entri Data Ahli

[Kembali](#)

Maklumat Akaun

Nama

Emel

Maklumat Ahli

No Anggota

No IC

No Telefon

Alamat

Bandar

Poskod

Negeri

Jantina
Pilih Jantina ▾

Tarikh Lahir
dd/mm/yyyy

Agama

Bangsa

Maklumat Pekerjaan

No PF

Gaji

Alamat Pejabat

Bandar Pejabat

Poskod Pejabat

Negeri Pejabat

Jawatan

Gred

Yuran dan Simpanan

Yuran Masuk

Modal Syer

Modal Yuran

Simpanan Ahli

Tabung Kebajikan

Simpanan Tetap

Maklumat Keluarga (Sila tambah ahli keluarga jika ada)

Bil.	Nama	No. KP	Hubungan	Tindakan
------	------	--------	----------	----------



Tiada ahli keluarga ditambah lagi

[+ Tambah Ahli Keluarga](#)

Maklumat Pinjaman (Pilihan)

☒ Tiada Pinjaman ☐ Ada Pinjaman

[Simpan Data Ahli](#)

Figure 4.1.5.5: Data Entry to transfer old member data to new system

KADA Admin

- Dashboard
- Senarai
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan
- Laporan Tahunan

Senarai Transaksi							
Pengurusan transaksi ahli							
Papar	10	entri	Cari:				
ID TRANSAKSI	NAMA AHLI	JENIS	JUMLAH (RM)	STATUS	TARIKH	TINDAKAN	
TRX202502160533582865	Poh Lok Yee	Bayaran Pinjaman	500.00	Diluluskan	16/02/2025 05:33		
TRX202502160537487880	Poh Lok Yee	Bayaran Pinjaman	4,000.00	Ditolak	16/02/2025 05:37		
TRX202502160804554183	Poh Lok Yee	Tabung Anggota	200.00	Tertunda	16/02/2025 08:04		
TRX202502160809123810	Poh Lok Yee	Bayaran Pinjaman	150.00	Tertunda	16/02/2025 08:09		
TRX202502160809482682	Poh Lok Yee	Bayaran Pinjaman	150.00	Diluluskan	16/02/2025 08:09		
TRX202502161122031767	Poh Lok Yee	Bayaran Pinjaman	1,000.00	Tertunda	16/02/2025 11:22		

Memaparkan 1 hingga 6 daripada 6 entri

Sebelumnya 1 Seterusnya

Figure 4.1.5.6: Transaction Management List

KADA Admin

- Dashboard
- Senarai
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan
- Laporan Tahunan

[← Kembali ke Senarai Transaksi](#)

Lulus Transaksi

Tolak Transaksi

Maklumat Transaksi

ID Transaksi	Status
TRX202502161122031767	Tertunda
Jenis Transaksi	Jumlah (RM)
Bayaran Pinjaman	1,000.00
Kaedah Pembayaran	Tarikh Transaksi
Online Banking	16/02/2025 11:22

Maklumat Ahli

Nama Penuh	No. KP
Poh Lok Yee	040923140332
Email	No. Telefon
lokjee1418@gmail.com	0111510260

Bukti Pembayaran

Maklumat Pinjaman

ID Pinjaman	Baki Pinjaman
LOAN-1739655096	RM 4,614.64
Jumlah Pinjaman Asal	
RM 5,000.00	

Figure 4.1.5.7: Member Transaction Application detail

KADA Admin

- Dashboard
- Senarai
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan**
- Laporan Tahunan

Senarai Pinjaman

Pengurusan pinjaman ahli

+ Kadar Faedah

Papar 10 entri

Cari:

ID PINJAMAN	NAMA PEMINJAM	JENIS PINJAMAN	JUMLAH (RM)	TEMPOH	STATUS	TINDAKAN
LOAN-1739655096	Poh Lok Yee	Al-Bai	5,000.00	24 bulan	Diluluskan	
LOAN-1739655234	Poh Lok Yee	Karnival Muslim Istimewa	50,000.00	24 bulan	Ditolak	
LOAN-1739664782	Lily	Skim Khas	68,000.00	30 bulan	Diluluskan	
LOAN-1739849639	TAN ZHI MING	Al-Inah	8,000.00	8 bulan	Tertunda	

Memaparkan 1 hingga 4 daripada 4 entri

Sebelumnya

1

Seterusnya

Figure 4.1.5.8: Member Loan Application Management

KADA Admin

- Dashboard
- Senarai
- Ahli
- Permohonan Baru
- Transaksi
- Kewangan**
- Laporan Tahunan

Kembali ke Senarai Pinjaman

Lulus Pinjaman

Tolak Pinjaman

Maklumat Pinjaman

ID Pinjaman

LOAN-1739849639

Status Pinjaman

Tertunda

Jenis Pinjaman

Al-Inah

Tarikh Mohon

2025-02-18 00:00:00

Jumlah Pinjaman

RM 8,000.00

Tempoh Pinjaman

8 bulan

Kadar Faedah

5%

Bayaran Bulanan

RM 1,018.84

Jumlah Bayaran

RM 8,150.72

Maklumat Peribadi

Nama Penuh

TAN ZHI MING

No. KP

041118011233

Email

bryanzhiming@gmail.com

No. Telefon

0195729461

Alamat Rumah

UTM JB

Bandar

JOHOR

Poskod

84040

Negeri

Johor

Maklumat Pekerjaan

Maklumat Bank

Figure 4.1.5.9: Member Loan Application detail

Dashboard

Senarai

Ahli

Permohonan Baru

Transaksi

Kewangan

Laporan Tahunan

Laporan Tahunan

Pengurusan laporan tahunan

Paparan Laporan

MUAT NAIK FAIL

Tajuk

Penerangan

Tahun

2025

Pilih Imej Muka Depan PNG, JPG sehingga 5MB

Pilih Fail PDF PDF sehingga 100MB

Figure 4.1.5.10: Annual Report Management

4.2 SQL Statement (DDL & DML)

The SQL Statement has been divided into 5 different modules done by each member, which is Member Management Module, Loan Management Module, Savings and Dividend Management Module, Reporting Module and Administrative Module.

4.2.1 Member Management Module

Guest Table:

```
CREATE TABLE Guest (  
id INT PRIMARY KEY,  
password VARCHAR (255) NOT NULL,  
name VARCHAR (255) NOT NULL,  
email VARCHAR (255) NOT NULL  
);
```

Insert Data into Guest Table

```
INSERT INTO Guest (id, password, name, email)  
VALUES (1, 'admin123', 'Admin', 'admin@example.com');
```

Delete Guest Record

```
DELETE FROM Guest  
WHERE id = 1;
```

Member Table:**Insert Data into Member table**

```
INSERT INTO Member (  
id, created_at, updated_at, no_anggota, name, email, ic, phone, address, city, postcode,  
state, gender, DOB, agama, bangsa, no_pf, salary, office_address, office_city,  
office_postcode, office_state, guest_id, status  
)  
VALUES  
(101, NOW(), NOW(), 'A001', 'John Doe', 'johndoe@gmail.com', '910101-01-1234',  
'0123456789', '123 Main St', 'City A', '12345', 'State A', 'M', '1991-01-01', 'Buddha', 'Cina',  
121, 5000, '123 Main St', 'City A', '12345', 'State A', 1, 'pending');
```

(102, NOW(), NOW(), 'A002', 'Jane Smith', ' janesmith@gmail.com ', '920202-02-2345', '0198765432', '456 Elm St', 'City B', '23456', 'State B', 'F', '1992-02-02', 'Buddha', 'Cina', 122, 5200, '123 Main St', 'City A', '12345', 'State A', 2, 'pending');
Update Status of a Member UPDATE Member SET status = 'approved' WHERE id = 101;
Admin Table: Create Admin Table <pre>CREATE TABLE Admin (id INT PRIMARY KEY, password VARCHAR (255) NOT NULL, name VARCHAR (255) NOT NULL, email VARCHAR (255) NOT NULL);</pre>
Insert Data into Admin Table <pre>INSERT INTO Admin (id, password, name, email) VALUES (1, 'admin123', 'Admin', 'admin@example.com');</pre>
Family Table: Create Family Table <pre>CREATE TABLE Family (id INT PRIMARY KEY, name VARCHAR(255) NOT NULL, ic VARCHAR(14) NOT NULL, relationship VARCHAR(50) NOT NULL, no_anggota VARCHAR (50) NOT NULL);</pre>
Insert Data into Family Table <pre>INSERT INTO Family (ID, name, ic, relationship,no_anggota) VALUES (1, 'David Doe', '910101-01-5555', 'Father', '101'), (2, 'Emily Doe', '920202-02-6666', 'Mother', '102');</pre>
StaffInfo Table:

Insert Data into working_info Table

```
INSERT INTO working_info (id, jawatan, gred, no_pf, salary, no_anggota)
VALUES (1, 'Manager', 'G5', '123', 5000, '101'),
       (2, 'Clerk', 'G2', '456', 5200, '102');
```

Savings Table:

Insert Data into Savings Table

```
INSERT INTO Savings (id, entrance_fee, share_capital, subscription_capital,
member_deposit, welfare_fund, fixed_savings, total_amount, no_anggota)
VALUES
  (1, 100.00, 1000.00, 200.00, 500.00, 50.00, 3000.00, 6000.00, 101),
  (2, 50.00, 500.00, 100.00, 300.00, 25.00, 1500.00, 3500.00, 102);
```

Join Queries

Query Member and Guest

```
SELECT Member.id AS member_id, Member.name, Guest.name AS guestUsername,
Guest.email AS guestEmail
FROM Member
INNER JOIN Guest ON Member.guest_id = Guest.id;
```

Query Member and Family

```
SELECT Member.member_id, Member.name, Guest.guestUsername, Guest.guestEmail
FROM Member
INNER JOIN Guest ON Member.guest_id = Guest.guest_id;
```

Query Member and Savings

```
SELECT Member.member_id, Member.name, Savings.savingAmount,
Savings.dividendRate
FROM Member
INNER JOIN Savings ON Member.member_id = Savings.member_id;
```

4.2.2 Loan Management Module

```
CREATE TABLE `loans` (
```

<pre> `loan_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL, `member_id` bigint UNSIGNED NOT NULL, `loan_type_id` bigint UNSIGNED NOT NULL, `bank_id` bigint UNSIGNED NOT NULL, `date_apply` date NOT NULL, `loan_amount` double NOT NULL, `interest_rate` double NOT NULL, `monthly_repayment` double NOT NULL, `monthly_gross_salary` double NOT NULL, `monthly_net_salary` double NOT NULL, `loan_period` int NOT NULL, `status` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'pending', `created_at` timestamp NULL DEFAULT NULL, `updated_at` timestamp NULL DEFAULT NULL) </pre>
<pre> CREATE TABLE `loan_types` (`loan_type_id` bigint UNSIGNED NOT NULL, `loan_type` varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL, `created_at` timestamp NULL DEFAULT NULL, `updated_at` timestamp NULL DEFAULT NULL) </pre>
<pre> INSERT INTO `member_register` (`id`, `created_at`, `updated_at`, `no_anggota`, `name`, `email`, `ic`, `phone`, `address`, `city`, `postcode`, `state`, `gender`, `DOB`, `agama`, `bangsa`, `no_pf`, `salary`, `office_address`, `office_city`, `office_postcode`, `office_state`, `guest_id`, `status`) VALUES (1, '2025-01-24 16:38:36', '2025-01-24 16:39:08', '1234', 'gui', 'guisin@graduate.utm.my', '020304020202', '0129715408', 's', 'tangkak', '84900', 'Johor', 'Perempuan', '2025-01-03', 'Buddha', 'Cina', '344', '8000', 'vv', 'kelantan', '12345', 'Kelantan', 2, 'approved'); </pre>
<pre> INSERT INTO `loans` (`loan_id`, `member_id`, `loan_type_id`, `bank_id`, `date_apply`, `loan_amount`, `interest_rate`, `monthly_repayment`, `monthly_gross_salary`, `monthly_net_salary`, `loan_period`, `status`) VALUES ('LOAN-1738035268', 1, 3, 1, '2025-01-28', 2888, 5, 247.23, 4000, 3600, 12, 'pending'); </pre>
<pre> UPDATE Loans SET status = 'approved' WHERE loan_id = 'LOAN-1738035268'; </pre>
<pre> DELETE FROM `loans` WHERE loan_id = 'LOAN-1738035268'; </pre>

```
SELECT loans.loan_id, member_register.name, loans.loan_amount, loans.status
FROM loans
JOIN member_register ON loans.member_id = member_register.id;
```

```
SELECT loans.loan_id, loan_types.loan_type, banks.bank_name, loans.loan_amount
FROM loans
JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
JOIN banks ON loans.bank_id = banks.bank_id
WHERE loans.status = 'approved';
```

```
SELECT loan_id, loan_amount, monthly_gross_salary
FROM loans
WHERE status = 'pending';
```

```
SELECT loan_types.loan_type, SUM(loans.loan_amount) AS total_loan
FROM loans
JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
GROUP BY loan_types.loan_type;
```

4.2.3 Savings and Dividend Management Module

```
CREATE TABLE Transactions (
    transaction_id INT PRIMARY KEY AUTO_INCREMENT
    type ENUM ('savings', 'loan') NOT NULL,
    savings_type VARCHAR (255) DEFAULT NULL,
    amount DECIMAL (10, 2) NOT NULL, member_id INT NOT NULL,
    FOREIGN KEY (MemberID) REFERENCES Savings(MemberID),
    FOREIGN KEY (LoanID) REFERENCES Loans(LoanID)
);
```

```
CREATE TABLE savings (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY, created_at
    TIMESTAMP NULL, updated_atTIMESTAMP NULL,
    entrance_fee DECIMAL (10, 2) NOT NULL,
    share_capital DECIMAL (10, 2) NOT NULL,
    subscription_capital DECIMAL (10, 2) NOT NULL,
    member_deposit DECIMAL (10, 2) NOT NULL,
```

welfare_fund DECIMAL (10, 2) NOT NULL, fixed_savings DECIMAL (10, 2) NOT NULL, total_amount DECIMAL (10, 2) NOT NULL, no_anggota BIGINT UNSIGNED NOT NULL
UPDATE Transactions SET savings_type = 'loan' WHERE id = 1;
DELETE FROM transactions WHERE savingstype = 'loan';
SELECT * FROM `transactions` WHERE savings_type = 'share capital';
INSERT INTO savings(entrance_fee, share_capital, subscription_capital, member_deposit, welfare_fund, fixed_savings, total_amount, no_anggota) VALUES (50.00,1000.00,200.00,1000.00,100.00,200.00,2550,1)
SELECT m.id, m.name, m.email, m.phone, m.address, m.city, m.postcode, m .state, m.gender, m.DOB, m.agama, m.bangsa, m.no_pf, m.salary, m.office_a ddress, m.office_postcode, m.office_state, m.no_anggota, s.total_amount FROM member_register m JOIN savings s ON m.id = s.id WHERE m.id =1;
SELECT m.member_id, m.member_name, SUM (CASE WHEN t.transactions_type = 'saving' THEN t.repayment_amount ELSE 0 END) AS total_savings_amount, SUM (CASE WHEN t.transactions_type = 'loan' THEN t.repayment_amount ELSE 0 END) AS total_loan_repayment_amount FROM transactions t JOIN

members m ON t.member_id = m.member_id GROUP BY m.member_id, m.member_name;
SELECT m.id AS member_id, m.name AS member_name, SUM (CASE WHEN t.type = 'savings' THEN t.amount ELSE 0 END) AS total_savings_amount, SUM (CASE WHEN t.type = 'loan' THEN t.amount ELSE 0 END) AS total_loan_repayment_amount FROM transactions t JOIN member_register m ON t.member_id = m.id GROUP BY m.id, m.name;
INSERT INTO transactions (type, savings_type, amount, member_id) VALUES ('savings', 'Simpanan Tetap', 1000.00, 1);

4.2.4 Reporting Module

CREATE TABLE IndividualReport (individualReport_id INT PRIMARY KEY, member_id INT NOT NULL, loan_id INT NOT NULL, savings_id INT NOT NULL, FOREIGN KEY (member_id) REFERENCES Member(member_id), FOREIGN KEY (loan_id) REFERENCES Loans(loan_id), FOREIGN KEY (savings_id) REFERENCES Savings(savings_id));
INSERT INTO IndividualReport (individualReport_id, member_id, loan_id, savings_id) VALUES (1, 101, 202, 303);
SELECT * FROM IndividualReport; WHERE member_id = 101;
SELECT m.name, l.loan_id, l.loan_amount, l.monthly_repayment, l.interest_rate, l.status FROM member_register m

JOIN loans l ON m.id = l.member_id WHERE m.id = 1;
SELECT * FROM loans WHERE member_id=1;
SELECT * FROM Member WHERE member_id=1;
SELECT m.name AS member_name, g.name AS guarantor_name, g.relationship, g.phone FROM Member m JOIN loans l ON m.id = l.member_id JOIN guarantor g ON l.loan_id = g.loan_id;
SELECT address, city, postcode, state, office_address, office_city, office_postcode, office_state FROM Member WHERE id=1;
CREATE TABLE AnnualReport (annualReport_id INT PRIMARY KEY AUTO_INCREMENT, filePDF VARCHAR(255) NOT NULL);
INSERT INTO AnnualReport (filePDF) VALUES ('/uploads/reports/annual2025.pdf');
UPDATE AnnualReport SET filePDF = '/uploads/reports/updated_annual2025.pdf' WHERE annualReport_id = 1;
DELETE FROM AnnualReport WHERE annualReport_id = 1;
SELECT * FROM AnnualReport ORDER BY annualReport_id DESC LIMIT 1;

```
SELECT COUNT(*) AS total_members  
FROM Member;
```

```
SELECT COUNT(*) AS members_in_process  
FROM Member  
WHERE status = 'pending';
```

```
SELECT SUM(total_amount) AS total_savings  
FROM Savings;
```

```
SELECT MONTH(date_apply) AS month, COUNT(*) AS total_approved_loans  
FROM loans  
WHERE YEAR(date_apply) = 2025  
AND status = 'approved'  
GROUP BY MONTH(date_apply);
```

```
SELECT MONTH(created_at) AS month, COUNT(*) AS total_new_members  
FROM Member WHERE YEAR(created_at) = 2025  
GROUP BY MONTH(created_at);
```

4.2.5 Administrative Module

SELECT: Get pending loan applications with member details for approval

```
SELECT l.loan_id,  
       m.name AS member_name,  
       m.ic AS member_ic,  
       lt.loan_type,  
       l.loan_amount,  
       l.monthly_gross_salary,  
       l.monthly_net_salary,  
       l.monthly_repayment,  
       l.date_apply,  
       b.bank_name,  
       b.bank_account  
FROM loans l  
JOIN member_register m ON l.member_id = m.id  
JOIN loan_types lt ON l.loan_type_id = lt.loan_type_id
```

<pre> JOIN banks b ON l.bank_id = b.bank_id WHERE l.status = 'pending' ORDER BY l.date_apply ASC; </pre>
<i>UPDATE: Approve a loan application</i> <pre> UPDATE loans SET status = 'approved', updated_at = CURRENT_TIMESTAMP WHERE loan_id = 'LOAN-1737940292' AND status = 'pending'; </pre>
<i>INSERT: Record loan disbursement transaction</i> <pre> INSERT INTO transactions (member_id, type, loan_id, amount, created_at, updated_at) SELECT l.member_id, 'loan', l.loan_id, 100.00, NOW(), NOW() FROM loans l WHERE l.loan_id = 'LOAN-1737940292' AND l.status = 'approved' AND l.member_id = 1; </pre>
<i>UPDATE: Approve member registration</i> <pre> UPDATE member_register SET status = 'approved', updated_at = CURRENT_TIMESTAMP WHERE id = 2 AND status = 'pending'; </pre>
<i>JOIN Query: Get all approved members with their active loans</i>


```

SELECT m.id, m.name, m.ic, m.email,
       COUNT(l.loan_id) as active_loans,
       SUM(l.loan_amount) as total_loan_amount
FROM member_register m
LEFT JOIN loans l ON m.id = l.member_id
AND l.status = 'approved'
WHERE m.status = 'approved'
GROUP BY m.id, m.name, m.ic, m.email
ORDER BY m.id;

```

JOIN Query: View transaction history at individual report

```

SELECT
  t.*,
  CASE
    WHEN t.type = 'savings' THEN 'Simpanan'
    WHEN t.type = 'loan' THEN 'Bayar Balik'
    ELSE t.type
  END as type_display,
  CASE
    WHEN t.type = 'loan' THEN CONCAT('LOAN-', l.loan_id)
    ELSE '-'
  END as reference
FROM transactions t
LEFT JOIN loans l ON t.loan_id = l.loan_id
WHERE t.member_id = 1
ORDER BY t.created_at DESC;

```

JOIN Query: Get loan approval statistics by loan status

```

SELECT lt.loan_type,
       COUNT(l.loan_id) as total_applications,
       SUM(CASE WHEN l.status = 'approved' THEN 1 ELSE 0 END) as
       approved_count,
       SUM(CASE WHEN l.status = 'rejected' THEN 1 ELSE 0 END) as rejected_count,
       AVG(CASE WHEN l.status = 'approved' THEN l.loan_amount ELSE 0 END) as
       avg_approved_amount
FROM loan_types lt
LEFT JOIN loans l ON lt.loan_type_id = l.loan_type_id
GROUP BY lt.loan_type_id, lt.loan_type
ORDER BY total_applications DESC;

```

SELECT: Get pending loan applications with member details for approval

```

SELECT l.loan_id,
       m.name AS member_name,
       m.ic AS member_ic,
       lt.loan_type,
       l.loan_amount,
       l.monthly_gross_salary,
       l.monthly_net_salary,
       l.monthly_repayment,
       l.date_apply,
       b.bank_name,
       b.bank_account
FROM loans l
JOIN member_register m ON l.member_id = m.id
JOIN loan_types lt ON l.loan_type_id = lt.loan_type_id
JOIN banks b ON l.bank_id = b.bank_id
WHERE l.status = 'pending'
ORDER BY l.date_apply ASC;

```

Create Transactions Table

```

CREATE TABLE transactions (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  member_id BIGINT UNSIGNED NOT NULL,
  type ENUM('savings', 'loan') NOT NULL,
  savings_type VARCHAR(255) NULL,
  loan_id VARCHAR(50) NULL,
  amount DECIMAL(10,2) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NULL,
  FOREIGN KEY (member_id) REFERENCES member_register(id),
  FOREIGN KEY (loan_id) REFERENCES loans(loan_id)
);

```

Create Loans Table

```

CREATE TABLE loans (
  loan_id VARCHAR(50) PRIMARY KEY,
  member_id BIGINT NOT NULL,
  loan_type_id BIGINT NOT NULL,
  bank_id BIGINT NOT NULL,
  date_apply DATE,
  loan_amount DECIMAL(10,2),
  interest_rate DECIMAL(5,2),
  monthly_repayment DECIMAL(10,2),
  monthly_gross_salary DECIMAL(10,2),

```

```
monthly_net_salary DECIMAL(10,2),  
loan_period INTEGER,  
status VARCHAR(20) DEFAULT 'pending',  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP,  
FOREIGN KEY (member_id) REFERENCES member_register(id),  
FOREIGN KEY (loan_type_id) REFERENCES loan_types(loan_type_id),  
FOREIGN KEY (bank_id) REFERENCES banks(bank_id)  
);
```

5.Summary

The KADA System was developed to streamline the member registration and loan application processes for organization. The core focus of this project was the design, implementation and optimization of a MySQL database, which serves as the foundation for strong, retrieving and managing essential system data. Using phpMyAdmin as the primary database management tool, the system ensures efficient and secure data handling while enhancing the overall operational effectiveness of the organization.

The database design followed the principles of relational database management systems (RDBMS) to maintain data integrity, consistency and scalability. The Entity-Relationship Diagram (ERD) was created to model the relationships between core entities, including members, guests, savings, family, loans and others attributes. The design process involved identifying primary keys, foreign keys and establishing relationships such as one-to-many and many-to-many to accurately represent real-world interactions within the organization.

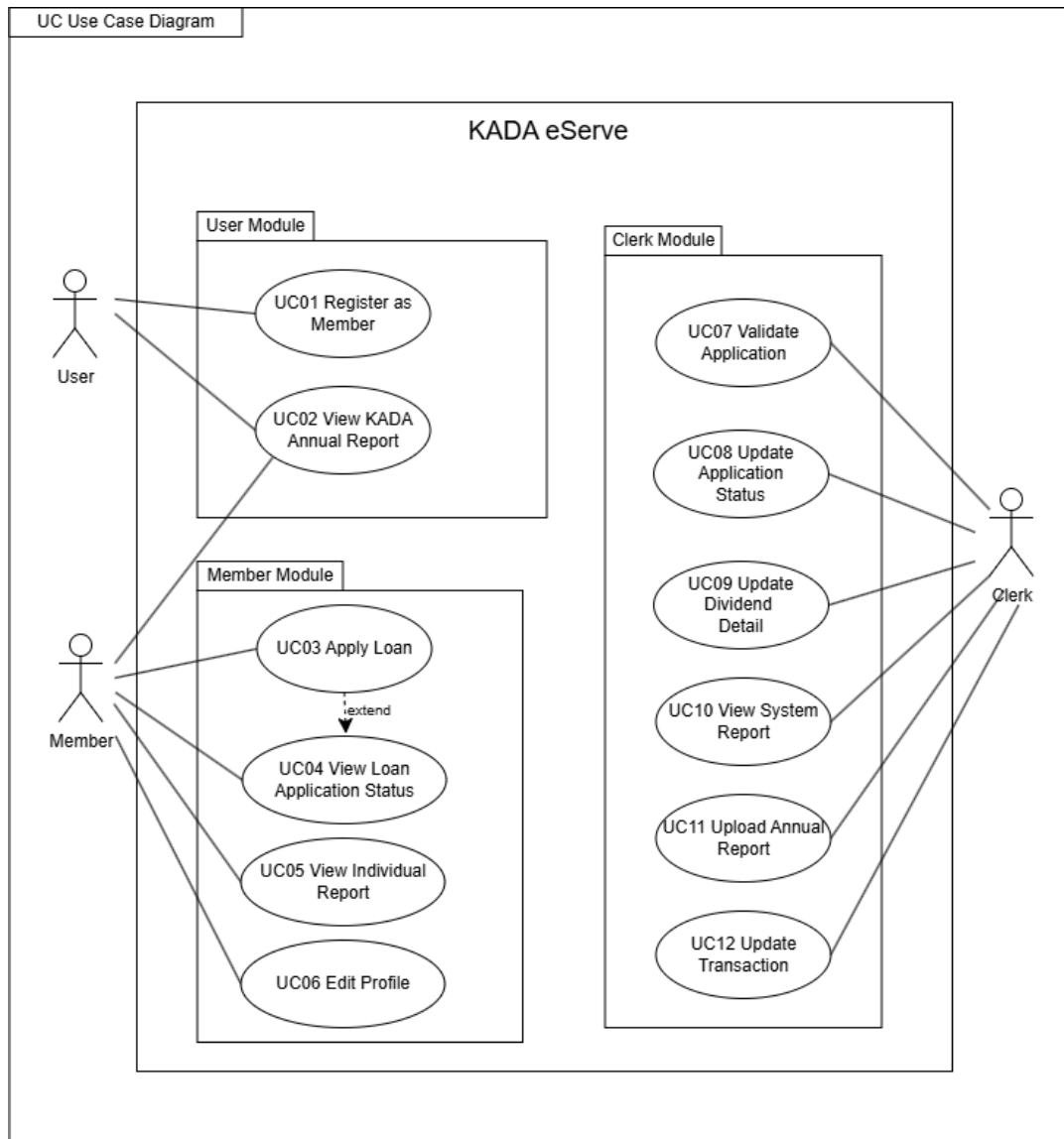
Normalization techniques were applied to the database schema to minimize data redundancy and ensure data integrity. Through normalization, the database was organized into several tables with clearly defined relationships. This structure not only optimizes storage but also improves the efficiency of queries and updates. Additionally, the use of constraints such as primary keys, foreign keys and unique indexes enforces data consistency across tables.

Security measures were incorporated to protect sensitive data within the database. Access control mechanisms, such as role-based access implemented via Bouncer, were employed to regulate database access based on user roles. Additionally, data validation techniques were applied to prevent SQL injection and other potential vulnerabilities. Regular backups were scheduled to ensure data availability and integrity in case of system failure.

In conclusion, the database design and implementation for the KADA System significantly enhanced the organization's ability to manage member-related data efficiently. The systematic approach to database modelling, normalization and security ensures a robust and scalable foundation for future enhancements. The successful integration of the database with the Laravel application demonstrates the critical role of database design in the system's overall performance and reliability.

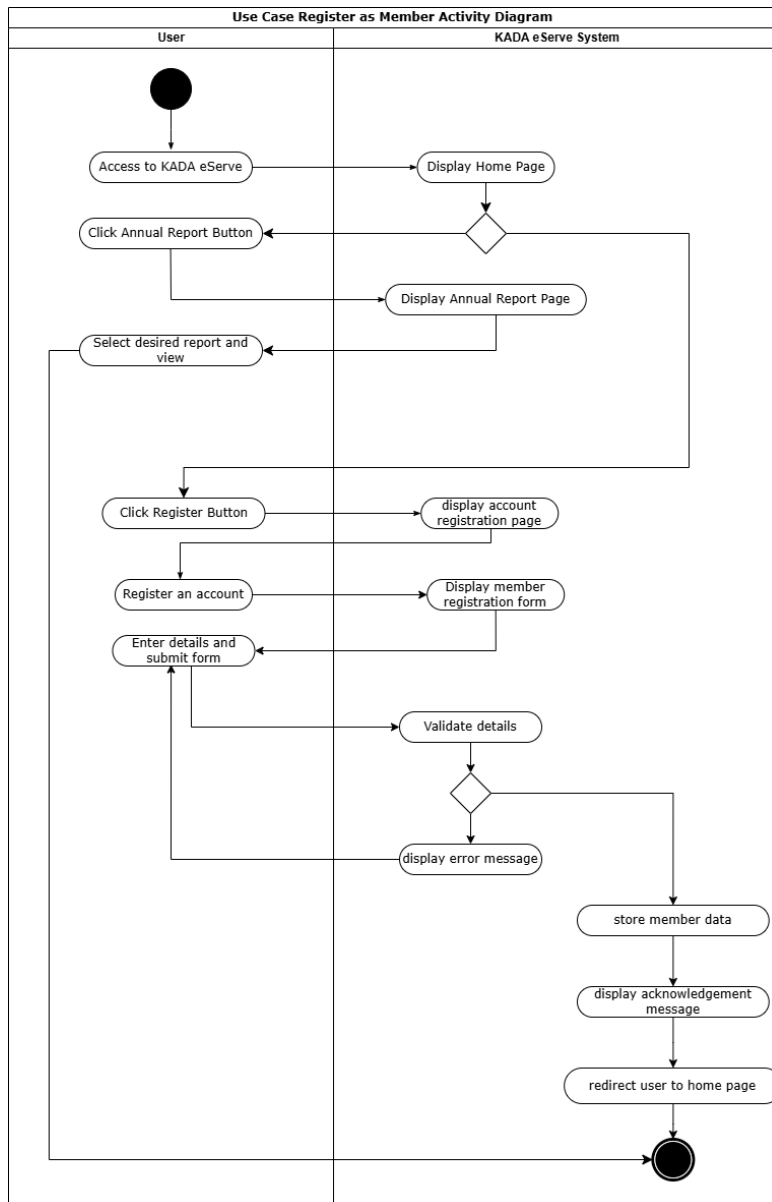
6. Appendices

6.1 Use Case Diagram for the system

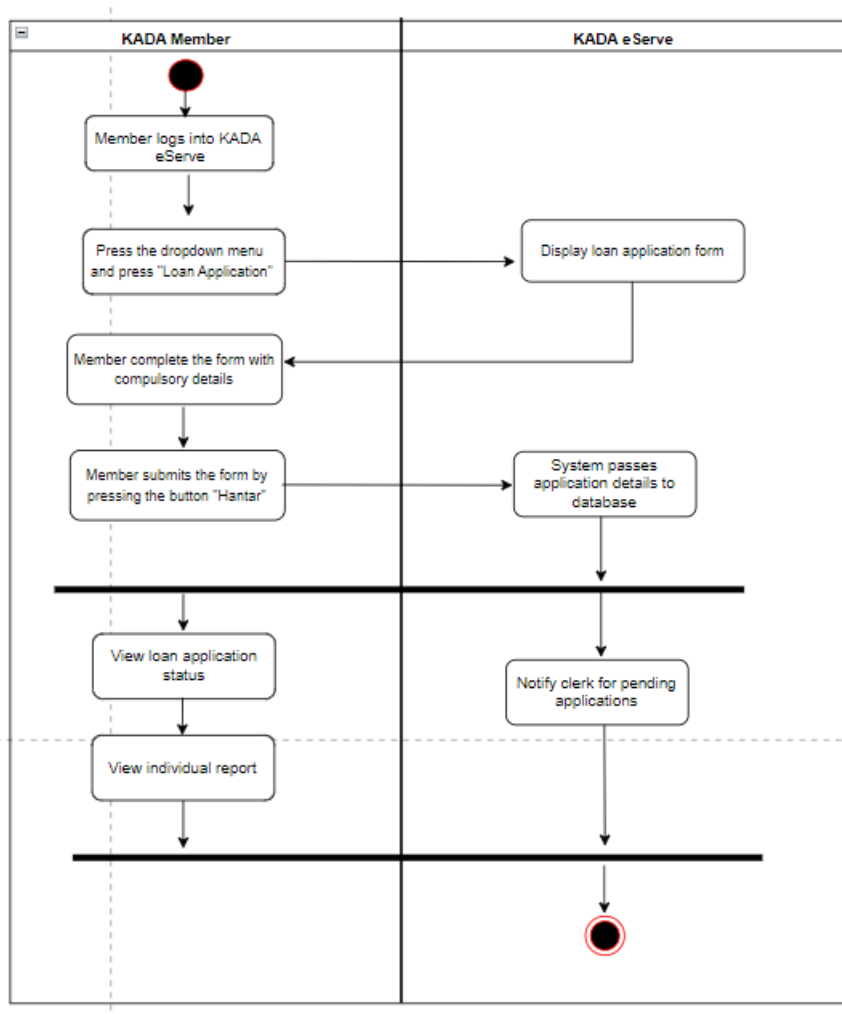


6.2 Activity Diagram for every Use Case identified

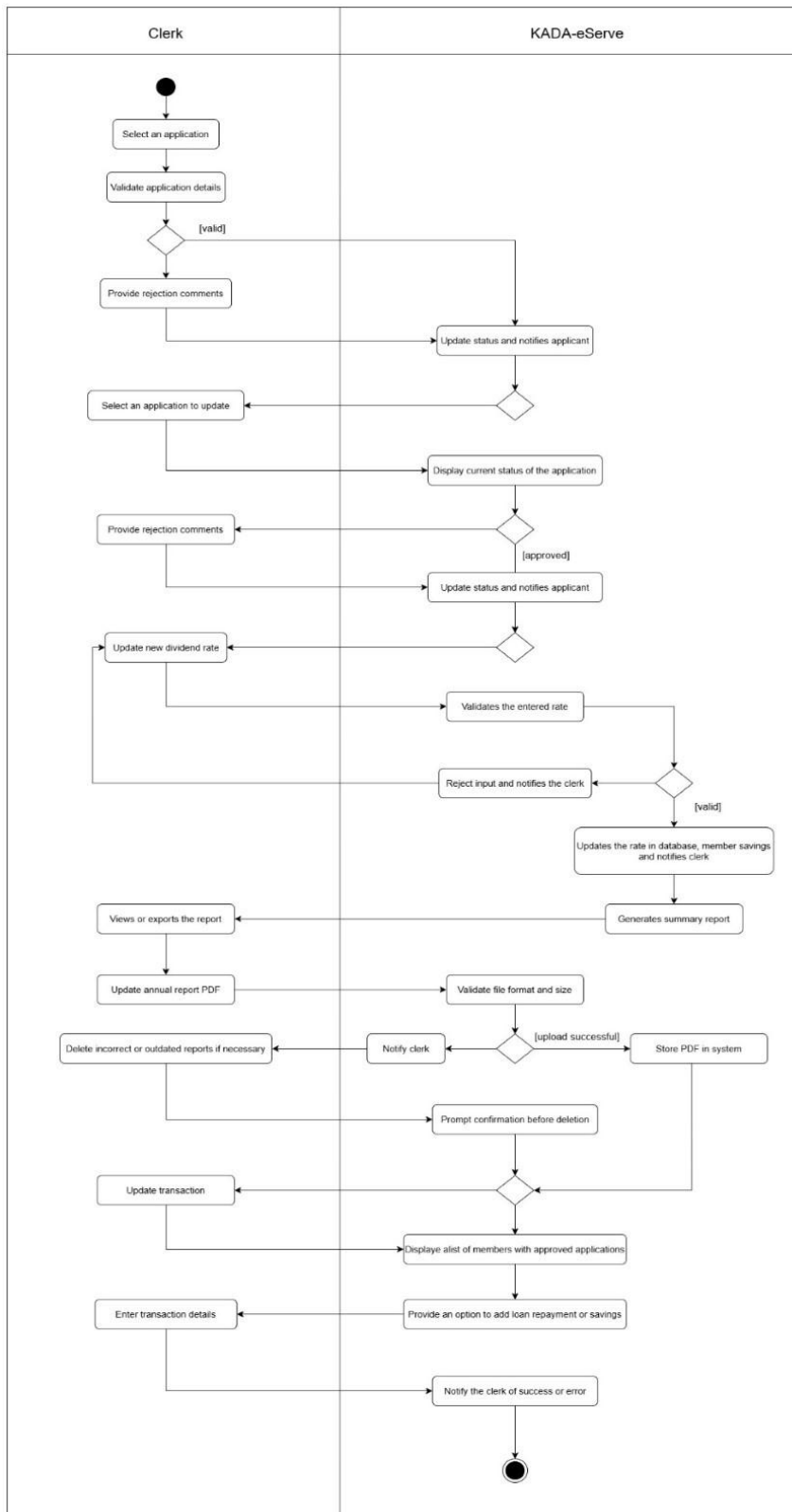
6.2.1 Module <User Module>



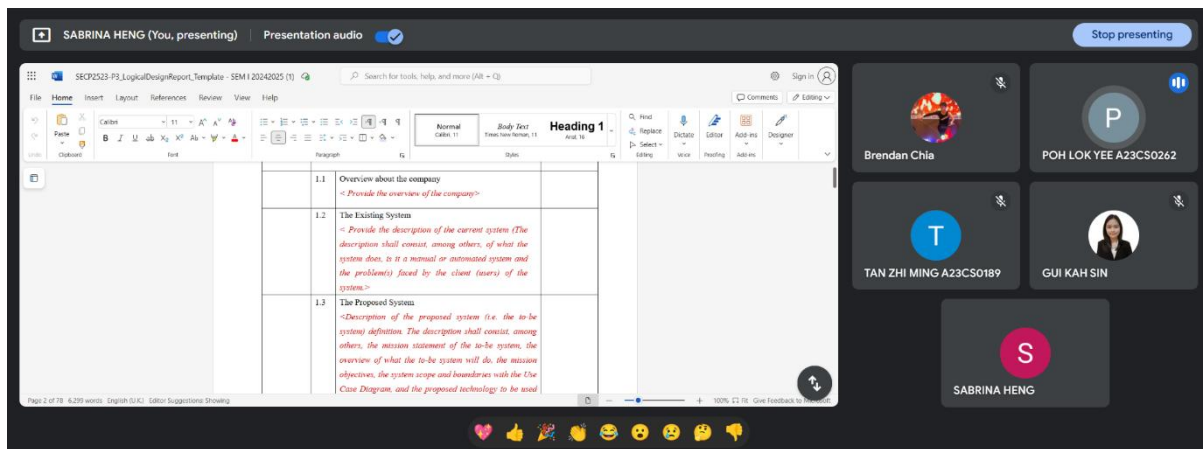
6.2.2 Module <Member Module>



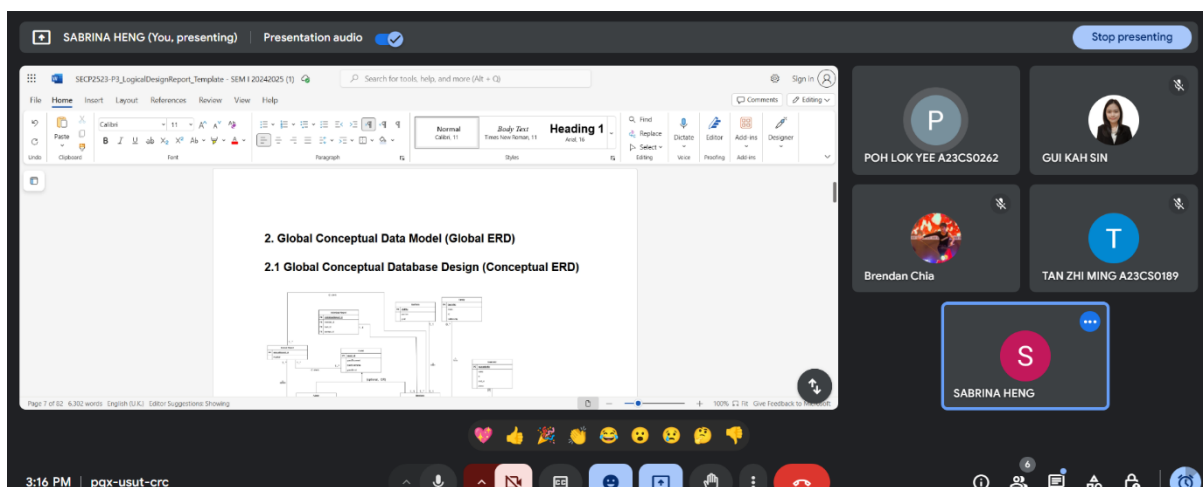
6.2.3 Module <Clerk Module>



6.3 Meeting Log



We start our meeting at 5:30 p.m. on 17 February 2025. In this meeting, the leader assign parts to each member and discussion about the third phase of the database project. Everyone attends the meeting punctually and having a smooth discussion time. Each member feels satisfied for the way of assign for their project parts. Lastly, this meeting is to have a brief discussion about the project and exchanging ideas about the ways to perform successfully in this project.



This meeting be conducted at 3p.m. on 19 January 2025 and the purpose of the meeting is to finalize each part of the project and discuss about the problem faced during the project. Each member updates their progress and speak out about issues found during this documentation. In the end of the meeting, we solved all of the problems and proceed to complete the report.